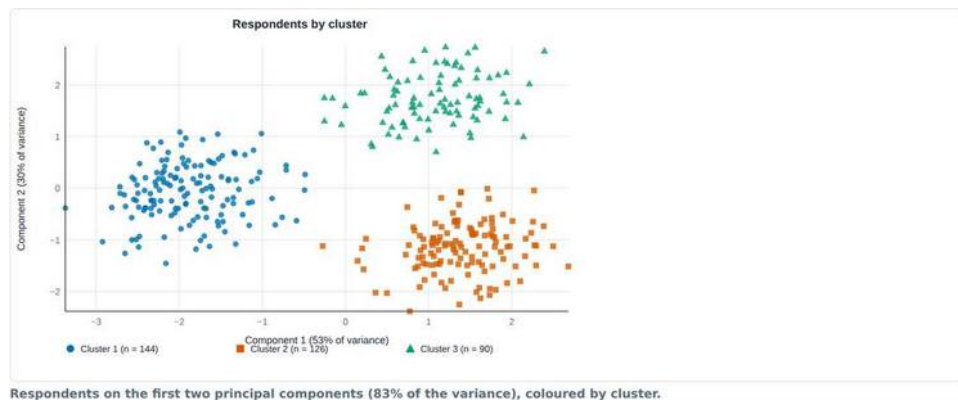


SaNaML

Marketing Analytics: a step-by-step manual with practice data

User manual • version 0.10.2 • October 2026



Worked examples: 1,500 simulated retail customers, a 500-respondent brand survey and 360 simulated shoppers for segmentation

Prof. S. Sabraz Nawaz and Prof. Ghazanfar Ali

SaNaSoft • 2026

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About this manual

This manual shows how to answer common marketing questions with SaNaML, the statistics and machine learning program of the SaNaSoft suite: who your customers are, how a brand survey scale behaves, whether a campaign worked, what drives spending, who is likely to leave, and which kinds of customers you have. Every chapter follows one practice file, tells you what to click, shows what you will see, explains how to read the result and gives a sentence for your report. All screenshots and numbers come from real runs of SaNaML 0.10.2, so you can follow along and obtain the same results.

Who this manual is for

Marketing students and researchers who have customer or survey data in a table (one row per customer or respondent, one column per variable) and want to analyse it carefully. You do not need earlier experience with any statistics software. Each statistical idea is explained in a "Concept" box when it first appears. Basic knowledge of means and percentages is enough to start. The manual complements the general SaNaML user manual, which describes every screen in detail; here we use only the parts that matter for marketing.

What SaNaML can and cannot do for marketing today

SaNaML 0.10.2 is a general program for statistics and machine learning. **It has no dedicated marketing menu.** Its screens are Home, Data, Analyses, Charts, Power analysis, Clustering, Machine learning and Word report. Marketing questions are answered by combining these general tools, and this manual shows how. We think it is better to say plainly where the limits are, so that you do not spend time looking for a button that does not exist.

Table 1

What the program does and does not offer for marketing analytics

Marketing task	Status in SaNaML 0.10.2
Describing customers, charts, correlation heat map	Available (chapter 2)
Making new variables (logarithm, difference of two columns, median split, 0/1 indicator, merged categories)	Available on the Data screen (sections 2.6, 2.7 and 4.3)
Survey scales: data quality screening, reliability (Cronbach's α), scale scores	Available (chapter 3)
A/B test of a response rate; before-and-after spend; sample size for a test	Available (chapter 4)
Drivers of spend (linear regression), group comparisons	Available (chapter 5)
Predicting churn or response (logistic regression, machine learning, explanations, calibration, cut-offs with precision and recall)	Available (chapter 6)
Customer segmentation (k-means and hierarchical clustering)	Available (chapter 7)

Marketing task	Status in SaNaML 0.10.2
RFM scoring (recency, frequency, monetary value)	Not available. You can compute the three columns elsewhere and bring them in as ordinary variables.
Customer lifetime value	Not available
Conjoint analysis and choice modelling	Not available
Association rules (market-basket analysis)	Not available
Time-series forecasting of sales or demand	Not available
Uplift modelling (which customers are moved by an offer)	Not available
Survival analysis of how long customers stay	Not available
Ordinal and multinomial regression	Not available

Note: what this means for you

If your project needs one of the unavailable methods, SaNaML can still help with the surrounding work: describing the data, checking its quality, testing differences between segments or campaign groups, and predicting an outcome. Use other tools for the method itself, and bring the resulting columns back into SaNaML as ordinary variables.

What you need

- A computer with an up-to-date browser. SaNaML runs inside the browser and your data stay on your computer; nothing is uploaded.
- Your data as a CSV, Excel (.xlsx) or SPSS (.sav / .zsav) file, with the variable names in the first row.
- The three practice files used here (see "The practice data" below). They ship with SaNaML.

The practice data

Table 2

The practice files used in this manual

File	What it contains	Used in
practice_marketing.csv	1,500 customers of a retailer: profile, shopping behaviour, satisfaction, a campaign test, spend and churn (20 variables)	Chapters 1, 2 and 4 to 8
practice_brand_survey.csv	500 respondents who rated Trust, Value and Loyalty items on a 1 to 7 scale, with age group, gender and customer type (15 variables)	Chapter 3
segments_demo.csv	360 shoppers rated on five attitude scales, with age, income, gender and channel	Chapter 7

All three files are simulated. No real customer or respondent appears in them. Effects were planted on purpose, so that you can check whether your analysis finds what was built in; chapters end with

exercises and the planted answer. The codebooks are in appendix A. Because the data are simulated, do not read the results as facts about real customers.

How to read the instructions

Table 3

Conventions used in this manual

Convention	Meaning	Example
Bold green text	A button, tab, menu item, box or list on the screen. Click or choose exactly this.	Run the comparison
Numbered steps	Actions to carry out in this order.	1. Open ... 2. Click ...
<i>Italics</i>	A technical term explained nearby.	<i>straight-lining</i>
Variable names	As they appear in the file.	Annual_spend

Concept: what a term means

Explains a statistical idea in plain language.

Cut-off values: the rule of thumb

The thresholds SaNaML applies, with their sources. SaNaML prints the same rule under every check ("Rule: ...").

What it means for our example

What the numbers of the worked example mean for the marketing question.

Tip

A shortcut or good practice.

Note

A warning or a point that needs judgement.

How to report it

A model sentence for your report. SaNaML writes one for every result.

Exercise

A task for you, with the planted answer where there is one.

1. Getting started with marketing data

Open SaNaML in your browser. The Home screen has the main menu: **Home**, **Data**, **Analyses**, **Charts**, **Power analysis**, **Clustering**, **Machine learning** and **Word report**. Every screen starts with a box *What is this screen? What do I do here?*, which explains the screen in two or three sentences.

1.1 Load the customer file

1. Click **Data** in the menu. Before any file is open, the screen shows the five steps for preparing data.
2. Click **Choose a file...** and pick *practice_marketing.csv*.
3. SaNaML reports "Open practice_marketing.csv: 1500 rows, 20 variables." (Figure 1).

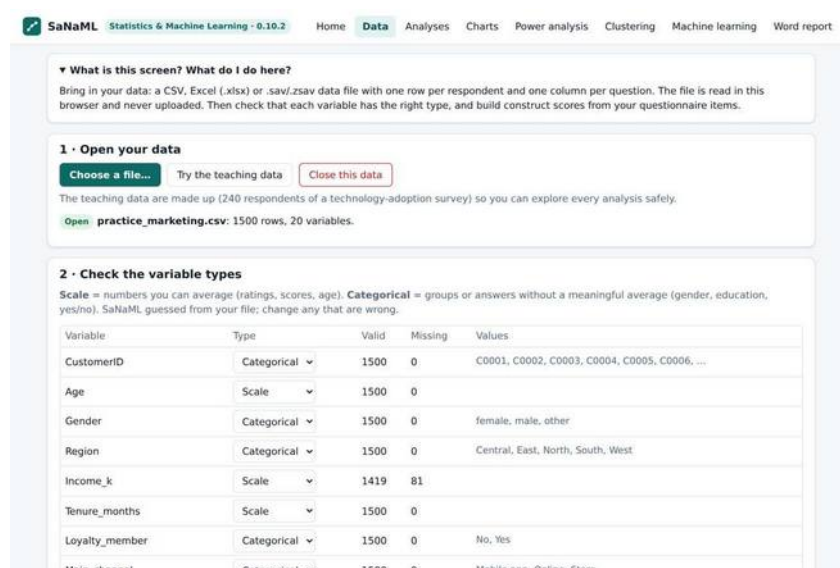


Figure 1. The Data screen after opening *practice_marketing.csv*.

Each row is one customer and each column one variable. The file is read inside the browser and is never uploaded. If you close the browser tab, SaNaML restores your data and results on your next visit in the same browser.

1.2 Check the variable types

Step 2 - *Check the variable types* lists every variable with its type, the number of valid and missing values, and, for categorical variables, the categories (Figure 2).

2 • Check the variable types

Scale = numbers you can average (ratings, scores, age). Categorical = groups or answers without a meaningful average (gender, education, yes/no). SaNaML guessed from your file; change any that are wrong.

Variable	Type	Valid	Missing	Values
CustomerID	Categorical	1500	0	C0001, C0002, C0003, C0004, C0005, C0006, ...
Age	Scale	1500	0	
Gender	Categorical	1500	0	female, male, other
Region	Categorical	1500	0	Central, East, North, South, West
Income_k	Scale	1419	81	
Tenure_months	Scale	1500	0	
Loyalty_member	Categorical	1500	0	No, Yes
Main_channel	Categorical	1500	0	Mobile app, Online, Store
Visits_per_month	Scale	1500	0	
Avg_basket	Scale	1500	0	
Discount_share	Scale	1500	0	
Complaints_12m	Scale	1500	0	
Satisfaction_1to10	Scale	1415	85	
Recommend_0to10	Scale	1435	65	
Annual_spend	Scale	1500	0	
Campaign_group	Categorical	1500	0	New offer, Standard
Responded	Categorical	1500	0	No, Yes
Monthly_spend_before	Scale	1500	0	
Monthly_spend_after	Scale	1500	0	
Churned	Categorical	1500	0	No, Yes

Figure 2. Step 2: the variable types of the customer file.

SaNaML distinguishes two types, and the type decides which analyses offer the variable.

- **Scale:** numbers that can be averaged, such as Age, Tenure_months, Annual_spend or a 1 to 10 rating.
- **Categorical:** groups or answers without a meaningful average, such as Region, Main_channel or Churned.

What it means for our example

SaNaML guessed correctly for most variables. Gender (female, male, other), Region (Central, East, North, South, West), Loyalty_member (No, Yes), Main_channel (Mobile app, Online, Store), Campaign_group (New offer, Standard), Responded (No, Yes) and Churned (No, Yes) are categorical. Age, Income_k, Tenure_months, Visits_per_month, Avg_basket, Discount_share, Complaints_12m, the two ratings, Annual_spend and the two monthly spend variables are scale variables.

Change any type that is wrong with the list in the **Type** column. A common marketing case is a rating such as Satisfaction_1to10: treat it as scale when you want means and regression, or as categorical when you want to compare the share of customers in each rating.

1.3 The ID column

CustomerID is a label, not a variable. SaNaML lists it as categorical, because every customer has a different value (C0001, C0002, and so on). **Do not use it as a predictor or a grouping variable.** In the analyses you simply never choose it. The Machine learning screen recognises a text identifier of this kind and leaves it unticked in the list of predictors (section 6.3).

1.4 Missing values

The **Missing** column shows blank cells. In `practice_marketing.csv` three variables have missing values: `Income_k` (81 missing, so 1419 valid), `Satisfaction_1to10` (85 missing, 1415 valid) and `Recommend_0to10` (65 missing, 1435 valid). All other variables are complete.

Most analyses leave out a customer only for the variables the analysis uses (a correlation uses every pair with both values; a regression uses customers complete on all of its variables). The result always states how many customers were used, for example "n = 1341 (159 respondents with missing values left out)" in the regression of section 5.2. Machine learning fills missing values in using the training customers only (section 6.3).

Note: missing values are information

Customers who skipped a satisfaction question may differ from those who answered it. SaNaML's data quality screen (shown for the survey in chapter 3) reports whether the missing answers look random (Little's test). Always report how many customers were left out of each analysis.

1.5 Skew

Money and counts are usually right-skewed: most customers spend a moderate amount and a few spend a great deal. Section 2.1 shows how to see the skew, and section 2.6 shows what to do about it with a new variable.

Exercise

Exercise: the file

Open `practice_marketing.csv` and find: (a) how many variables are categorical; (b) which three variables have missing values and how many; (c) a variable that should not be used as a predictor.

Answer: (a) seven (Gender, Region, Loyalty_member, Main_channel, Campaign_group, Responded, Churned) plus CustomerID, which SaNaML also lists as categorical, so eight in the list; (b) `Income_k` 81, `Satisfaction_1to10` 85, `Recommend_0to10` 65; (c) CustomerID.

2. Know your customers

Before testing anything, look at the data. This chapter uses the **Analyses** screen (descriptive statistics) and the **Charts** screen.

2.1 Descriptive statistics

1. Click **Analyses** in the menu, then **Descriptive statistics** in the list on the left.
2. Tick the scale variables to describe: Age, Income_k, Tenure_months, Visits_per_month, Avg_basket, Discount_share and Annual_spend.
3. Read the table (Figure 3).

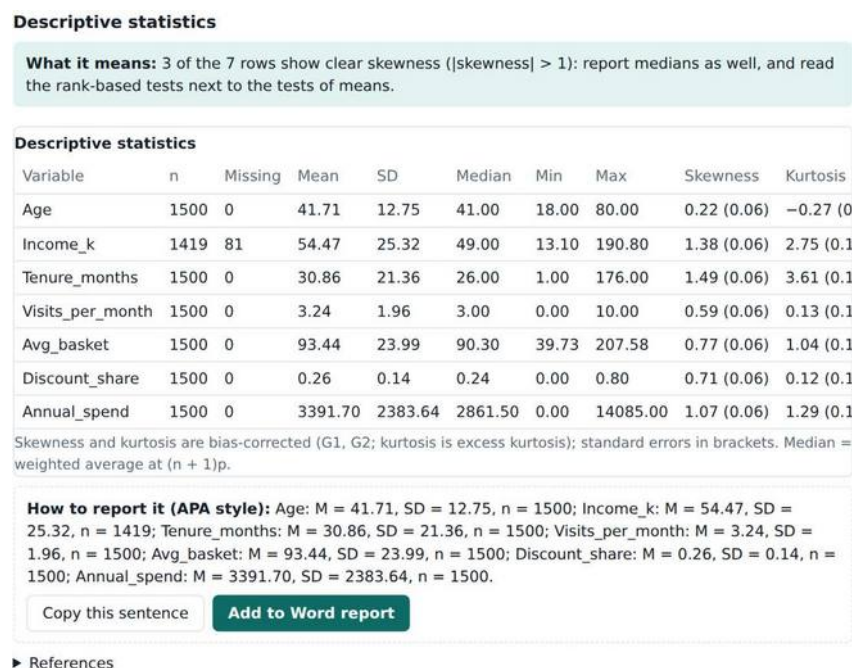


Figure 3. Descriptive statistics of seven customer variables.

Table 4

Descriptive statistics of the customers (n = 1500 unless stated)

Variable	Mean	SD	Median	Min	Max	Skewness
Age	41.71	12.75	41.00	18.00	80.00	0.22
Income_k (n = 1419)	54.47	25.32	49.00	13.10	190.80	1.38
Tenure_months	30.86	21.36	26.00	1.00	176.00	1.49
Visits_per_month	3.24	1.96	3.00	0.00	10.00	0.59
Avg_basket	93.44	23.99	90.30	39.73	207.58	0.77
Discount_share	0.26	0.14	0.24	0.00	0.80	0.71
Annual_spend	3391.70	2383.64	2861.50	0.00	14085.00	1.07

SaNaML summarises the result in its *What it means* line: "3 of the 7 rows show clear skewness ($|skewness| > 1$): report medians as well, and read the rank-based tests next to the tests of means."

Concept: mean, median and skewness

The *mean* is the arithmetic average; the *median* is the middle value. For a symmetric variable they are close. A long tail of large values (*positive skew*) pulls the mean above the median. In the table, the mean of Annual_spend (3391.70) is well above the median (2861.50), and its skewness is 1.07. SaNaML flags a variable as clearly skewed when the absolute skewness exceeds 1. For skewed money variables, report the median next to the mean.

Choose a categorical variable in **Split by** to describe each group. Splitting by Loyalty_member (Figure 4) shows what loyalty members do differently.

Descriptive statistics by Loyalty_member

What it means: 5 of the 14 rows show clear skewness ($|\text{skewness}| > 1$): report medians as well, and read the rank-based tests next to the tests of means.

Descriptive statistics

Variable	Loyalty_member	n	Missing	Mean	SD	Median	Min	Max	Skev
Age	No	854	0	41.64	12.88	41.00	18.00	78.00	0.25
Age	Yes	646	0	41.80	12.59	42.00	18.00	80.00	0.18
Income_k	No	811	0	54.89	25.49	50.50	13.10	190.80	1.38
Income_k	Yes	608	0	53.92	25.11	47.90	15.70	182.70	1.38
Tenure_months	No	854	0	28.50	19.84	23.00	2.00	153.00	1.60
Tenure_months	Yes	646	0	33.99	22.86	29.00	1.00	176.00	1.34
Visits_per_month	No	854	0	2.65	1.67	2.00	0.00	8.00	0.60
Visits_per_month	Yes	646	0	4.01	2.05	4.00	0.00	10.00	0.37
Avg_basket	No	854	0	88.77	24.01	86.00	39.73	207.58	0.95
Avg_basket	Yes	646	0	99.61	22.53	96.15	48.28	204.60	0.73
Discount_share	No	854	0	0.26	0.14	0.24	0.00	0.80	0.69
Discount_share	Yes	646	0	0.26	0.15	0.24	0.00	0.76	0.74
Annual_spend	No	854	0	2637.02	1893.02	2330.50	0.00	12460.00	1.12
Annual_spend	Yes	646	0	4389.38	2591.22	3949.00	0.00	14085.00	0.79

Skewness and kurtosis are bias-corrected (G1, G2; kurtosis is excess kurtosis); standard errors in brackets. Median = weighted average at $(n + 1)p$.

How to report it (APA style): Age: Loyalty_member = No, M = 41.64, SD = 12.88, n = 854; Loyalty_member = Yes, M = 41.80, SD = 12.59, n = 646. Income_k: Loyalty_member = No, M = 54.89, SD = 25.49, n = 811; Loyalty_member = Yes, M = 53.92, SD = 25.11, n = 608. Tenure_months: Loyalty_member = No, M = 28.50, SD = 19.84, n = 854; Loyalty_member = Yes, M = 33.99, SD = 22.86, n = 646. Visits_per_month: Loyalty_member = No, M = 2.65, SD = 1.67, n = 854; Loyalty_member = Yes, M = 4.01, SD = 2.05, n = 646. Avg_basket: Loyalty_member = No, M = 88.77, SD = 24.01, n = 854; Loyalty_member = Yes, M = 99.61, SD = 22.53, n = 646. Discount_share: Loyalty_member = No, M = 0.26, SD = 0.14, n = 854; Loyalty_member = Yes, M = 0.26, SD = 0.15, n = 646. Annual_spend: Loyalty_member = No, M = 2637.02, SD = 1893.02, n = 854; Loyalty_member = Yes, M = 4389.38, SD = 2591.22, n = 646.

Copy this sentence

Add to Word report

► References

Figure 4. Descriptive statistics split by Loyalty_member.

What it means for our example

Loyalty members visit more often (mean 4.01 visits per month against 2.65 for non-members), fill larger baskets (99.61 against 88.77) and spend more per year (mean 4389.38 against 2637.02). Their age and income are almost the same (mean age 41.80 and 41.64). Note that this is a description of two groups, not proof that the loyalty programme causes the extra spending: customers who spend more may be the ones who join.

How to report it

"Customers spent a mean of 3391.70 per year (SD = 2383.64, Mdn = 2861.50, n = 1500); the distribution was right-skewed (skewness = 1.07)."

(The sentence SaNaML itself writes lists M, SD and n for each variable, and for each group when you split the table; add the median and the skewness from the table when a variable is skewed.)

2.2 Histograms

A histogram shows how the values are spread. Click **Charts**, choose **Histogram (How are the values spread?)** as the **Chart type**, and choose **Annual_spend** as the **Variable** (Figure 5).



Figure 5. Histogram of Annual_spend, with the mean and median marked.

SaNaML states the finding: "Annual_spend has 1500 values with mean 3391.70, median 2861.50 and SD 2383.64. The distribution is skewed to the right (a long tail of high values)." The histogram of Income_k has the same shape (skewness 1.38).

Every chart has a short caption and a **How to report it** sentence, and an **Add to Word report** button.

2.3 Box plots: medians and spread by group

Choose **Box plot (Medians and spread, by group)**, **Annual_spend** as the variable and **Loyalty_member** as the groups (Figure 6). The box covers the middle half of the customers, the line is the median, the whiskers reach the values within 1.5 times the interquartile range and the circles are unusual values.

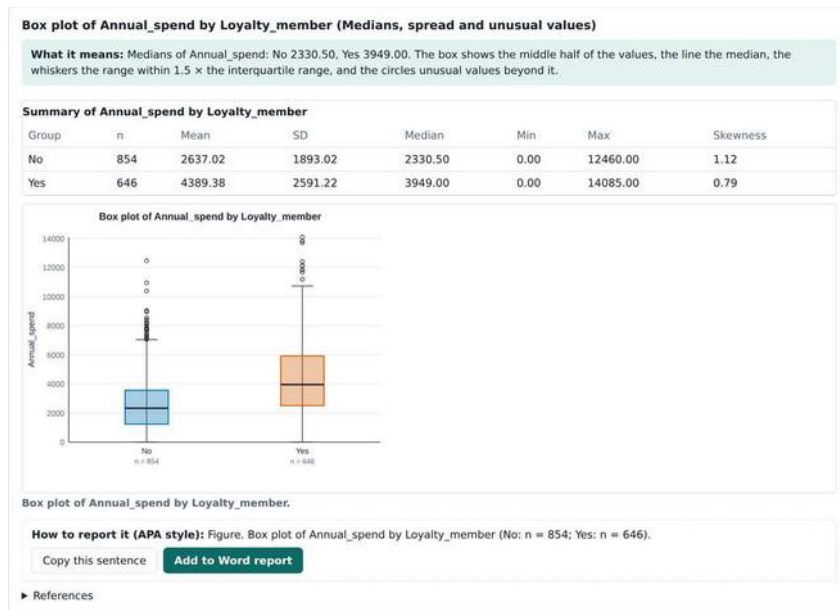


Figure 6. Box plot of Annual_spend by Loyalty_member.

Medians of Annual_spend: No 2330.50, Yes 3949.00. The same plot reveals a feature you cannot see in a table: the lowest values of both groups reach 0.00, which means that some customers spent nothing in the year (section 2.6).

2.4 Group means with confidence intervals

Choose **Means with confidence intervals (Group averages)**, the variable **Satisfaction_1to10** and the groups **Main_channel** (Figure 7).

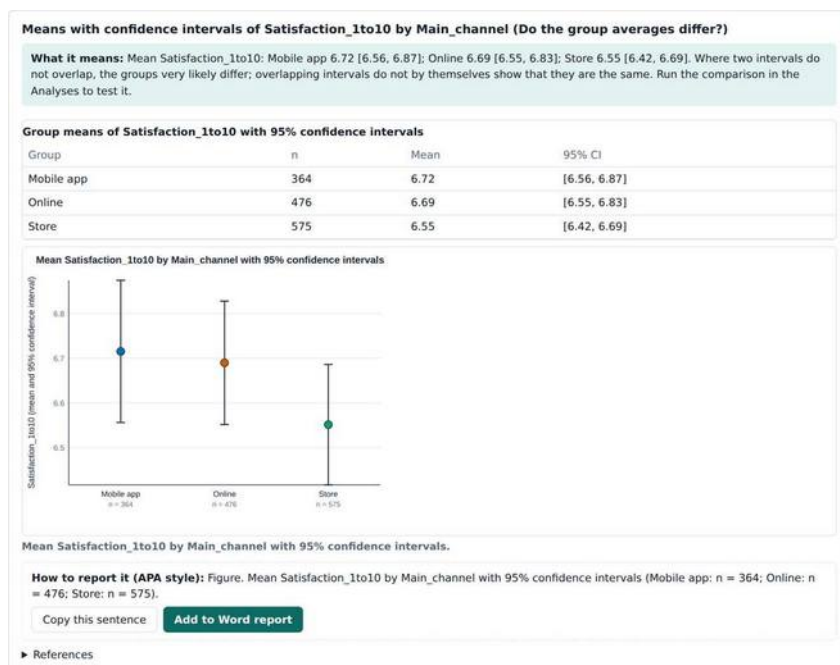


Figure 7. Mean satisfaction by main channel, with 95% confidence intervals.

Table 5

Mean Satisfaction_1to10 by Main_channel

Channel	n	Mean	95% CI
Mobile app	364	6.72	[6.56, 6.87]
Online	476	6.69	[6.55, 6.83]
Store	575	6.55	[6.42, 6.69]

Concept: confidence interval

A 95% confidence interval is a range of plausible values for the average of all customers like these, given the sample. Where two intervals do not overlap, the groups very likely differ. Overlapping intervals do not by themselves show that the groups are the same, and the chart is no substitute for a test; section 5.5 tests the same comparison.

What it means for our example

The three intervals overlap heavily: satisfaction is about 6.6 to 6.7 in every channel. Charts show patterns, not proof.

2.5 Scatter plots and the correlation heat map

Scatter plot. Choose **Scatter plot with line (Do two variables go together?)**, **Income_k** on the horizontal axis and **Annual_spend** on the vertical axis (Figure 8). SaNaML draws the dots and a dashed least-squares line and reports r . Optionally colour the dots by a group.

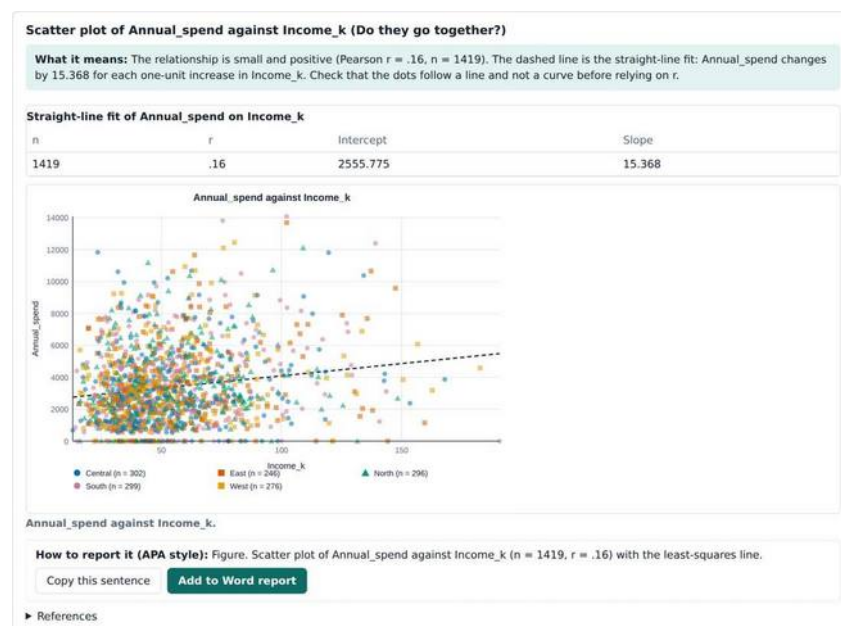


Figure 8. Scatter plot of Annual_spend against Income_k, coloured by region.

Higher income goes with slightly higher spend: "The relationship is small and positive (Pearson $r = .16$, $n = 1419$)", with a slope of 15.368 per extra thousand of income. The dots form a broad cloud around the line.

Correlation heat map. Choose **Correlation heatmap (Many variables at once)**, click **Select none**, and tick the nine variables Income_k, Tenure_months, Visits_per_month, Avg_basket, Discount_share, Complaints_12m, Satisfaction_1to10, Recommend_0to10 and Annual_spend (Figure 9). Red cells are positive correlations and blue cells negative ones; each cell shows r .

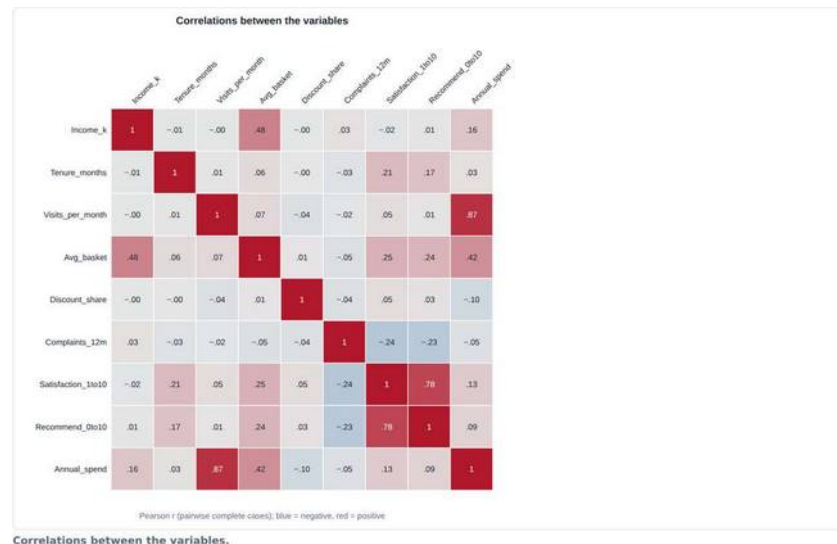


Figure 9. Correlation heat map of nine customer variables (Pearson r , pairwise complete cases).

What it means for our example

The strongest pair is Visits_per_month and Annual_spend ($r = .87$). Next come Satisfaction_1to10 and Recommend_0to10 ($r = .78$) and Income_k and Avg_basket ($r = .48$). Annual_spend also goes with Avg_basket ($r = .42$) and weakly with Income_k ($r = .16$). Complaints_12m is negatively related to satisfaction ($r = -.24$) and to willingness to recommend ($r = -.23$). Tenure_months goes with satisfaction ($r = .21$).

SaNaML adds: "Pairs above about .80 (in either direction) measure nearly the same thing, which matters when both are used as predictors." Satisfaction and the recommendation rating are close to that line, so a model should not be expected to separate their effects.

Note: when two variables are parts of a third

Annual spend is, to a good approximation, the number of visits multiplied by the value of the basket. A strong correlation between spend and visits is therefore partly arithmetic, not a discovery about what *drives* spending. Chapter 5 returns to this.

2.6 Category percentages and the problem of skew

Choose **Bar chart of percentages (Categories)** and a categorical variable. For Main_channel: Store 40.7% (611 customers), Online 34.1% (511) and Mobile app 25.2% (378). For Churned: No 83.3%

(1250) and Yes 16.7% (250). For Region: South 21.7%, Central 21.5%, North 20.5%, West 19.0% and East 17.3%.

The churn figure matters later: only about one customer in six leaves, so churn is an *imbalanced* outcome (chapter 6).

Annual_spend: skew, zeros and the logarithm

Section 2.2 showed that Annual_spend is right-skewed. A regression of a skewed variable gives residuals (prediction errors) that are not symmetric. The usual remedy for money is the *logarithm*, which turns multiplicative differences into additive ones and pulls in the long right tail. SaNaML makes the new variable for you, in step 4 of the **Data** screen, *Compute and recode variables (optional)*. The original variable is never changed; the new one is added to the list.

1. On the **Data** screen scroll to step 4 · *Compute and recode variables (optional)*.
2. In **What do you want to make?** keep **Transform a scale variable (logarithm, standardise, reverse ...)**, choose **Annual_spend** as the **Variable** and **$\ln(1 + x)$** as the **Transformation** (Figure 10).
3. Leave the name as suggested, **Annual_spend_ln1p**, and click **Add the new variable**. SaNaML confirms: "Added Annual_spend_ln1p: 1500 of 1500 respondents have a value."
4. Repeat with the transformation **Natural logarithm, $\ln(x)$** . The suggested name is **Annual_spend_ln**, and SaNaML warns: "Added Annual_spend_ln: 1418 of 1500 respondents have a value. Some values could not be computed and are missing (for example a logarithm needs positive values and a ratio needs a denominator other than 0)."

Figure 10. Step 4 of the Data screen, ready to add $\ln(1 + \text{Annual_spend})$.

The two choices differ in what they do with a spend of 0. The logarithm $\ln(x)$ needs a positive value, so the 82 customers who spent nothing get no value ($1500 - 82 = 1418$ remain). The version $\ln(1 + x)$ allows zeros and keeps all 1500 customers. Both new variables are scale variables. Now draw the histograms of the two.

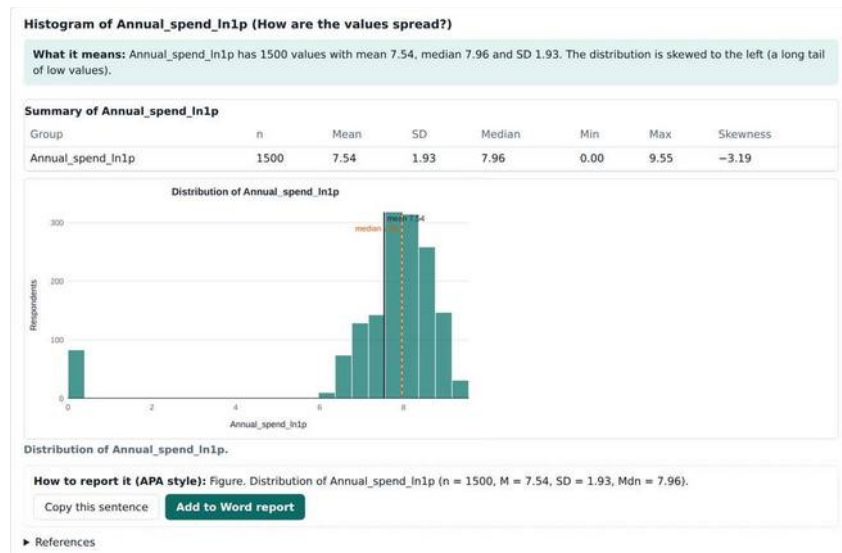


Figure 11. Histogram of Annual_spend_In1p (logarithm of one plus Annual_spend).

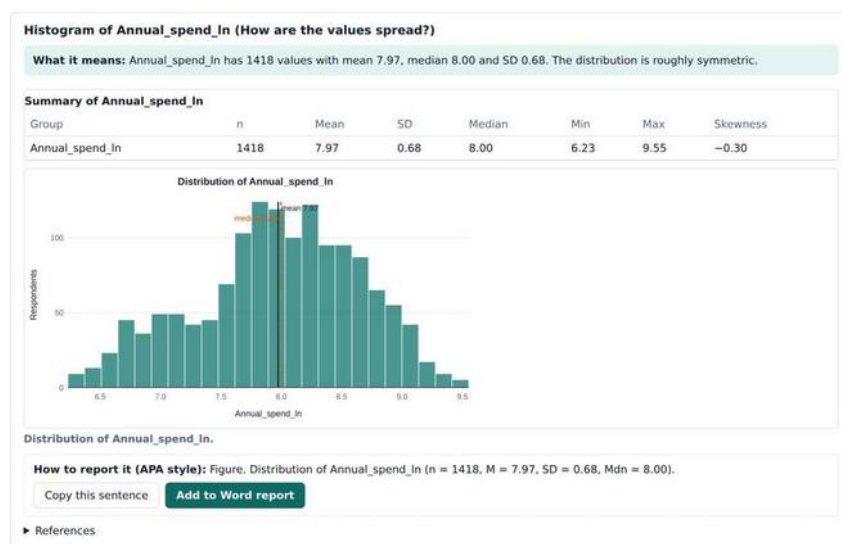


Figure 12. Histogram of Annual_spend_In (logarithm of Annual_spend; customers with zero spend have no value).

What it means for our example

Annual_spend_In1p is *not* better than the raw variable: its skewness is -3.19, a long tail to the left (mean 7.54, median 7.96, SD 1.93). The reason is that 82 customers spent nothing (Annual_spend = 0). A logarithm of one plus zero is zero, far below the next value (the smallest positive spend gives a logarithm of about 6.2), and this creates the left tail. Annual_spend_In leaves those 82 customers out (1418 values remain) and is nearly symmetric: mean 7.97, median 8.00, SD 0.68, skewness -0.30.

Note: choose the transformation with the question in mind

Customers with zero spend are real customers, probably the ones most at risk of leaving. Leaving them out of a logarithmic analysis answers a different question ("among customers who bought something, what drives how much?"). Say so in your report. Section 5.2 compares both versions.

2.7 Groups, indicators and merged categories

The same step 4 makes three other kinds of new variable. All of them leave the original untouched.

Two or more groups from a scale variable. Choose **Make groups from a scale variable (median split, tertiles ...)**, the variable **Satisfaction_1to10** and **Two groups at the median** (Figure 13). SaNaML shows the cut point before you add the variable: "Cut point: 6.6". A value equal to the cut point goes to the *lower* group. The new categorical variable is **Satisfaction_1to10_group**, with the groups *Low* ($\leq$ median) and *High* ($>$ median); 1415 of the 1500 customers have a value, because 85 did not answer the satisfaction question. The other choices are three equal-sized groups (**Three equal-sized groups (tertiles)**), four equal-sized groups (quartiles) and **My own cut points**: tertiles of **Income_k**, for example, have the cut points 40.4 and 59.733 and the groups *Low third*, *Middle third* and *High third*.

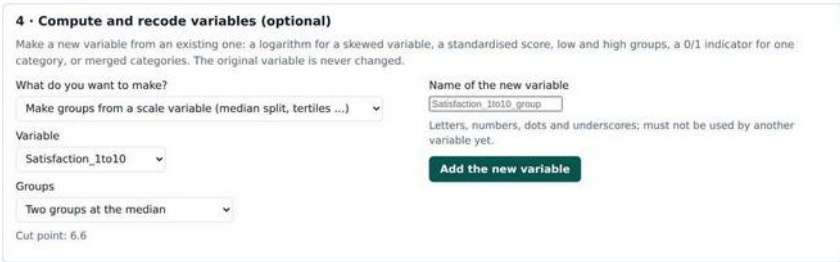


Figure 13. Making low and high satisfaction groups at the median (cut point 6.6).

A 0/1 indicator of a category. Choose **Make a 0/1 indicator of a category**, the variable **Churned** and tick **Yes**. The new scale variable **Churned_Yes** is 1 for a customer who churned and 0 for the others (1500 of 1500 customers have a value). Its mean is the share of customers who churned, 16.7% (250 of 1500).

Merging categories. Choose **Merge categories into one**, the variable **Region**, tick **East** and **West** and name the merged category **East_West**. The new variable is **Region_merged**, with the categories Central, East_West, North and South. The merged categories are counted together in every later analysis.

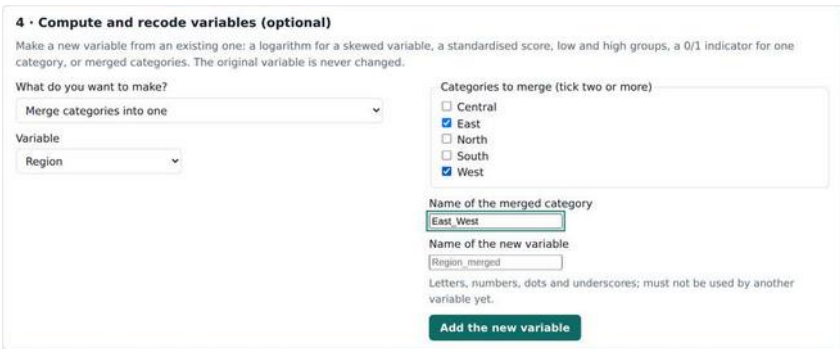


Figure 14. Merging East and West into one category.

What it means for our example

Customers with low satisfaction (at or below 6.6) churn far more often than those with high satisfaction: 25% (184 of 726) against 7% (50 of 689). The chi-square test of Satisfaction_1to10_group by Churned gives $\chi^2(1, N = 1415) = 83.79$, $p < .001$, Cramér's $V = .24$ (small). After merging East and West, churn is 23% (75 of 322) in Central, 15% (82 of 544) in East_West, 14% (44 of 308) in North and 15% (49 of 326) in South: $\chi^2(3, N = 1500) = 13.06$, $p = .005$, Cramér's $V = .09$ (negligible). Only Central has more churners than expected (adjusted residual 3.60).

Note: dichotomising a scale loses information

A median split throws away the differences within each group, and a customer just below the cut point is treated like one far below it. Use groups when the question is about groups, as in a report for managers; for a statistical analysis keep the scale variable where you can (the regression of section 5.2 uses Satisfaction_1to10 itself).

Combining two scale variables. Choose **Combine two scale variables (difference, change, ratio ...)**, the first variable (**Variable A**), the way to combine (**Difference, $A - B$, Sum, $A + B$, Product, $A \times B$, Ratio, $A \div B$ or Percentage change, $(A - B) \div B \times 100$) and the second variable (**Variable B**). A customer who has no value on A or on B gets no value, and neither does a customer with $B = 0$ when you ask for a ratio or a percentage change. Section 4.3 uses a difference.**

Exercises

Exercise: know your customers

1. Describe Tenure_months and Complaints_12m. Which of the two is more skewed? Answer: Tenure_months (skewness 1.49) against Complaints_12m (1.15), and both are flagged as clearly skewed.
1. Draw the box plot of Visits_per_month by Loyalty_member. Do members visit more often? Answer: yes, the group means are 4.01 and 2.65 (section 2.1).
1. Draw the heat map and find the pair of variables with the weakest correlation with Annual_spend. Answer: Complaints_12m ($r = -.05$), Tenure_months ($r = .03$) and Recommend_0to10 ($r = .09$) have almost no linear relation with spend; Discount_share is weakly negative ($r = -.10$).

3. Survey scales: trust, value and loyalty

Brand and customer surveys measure ideas such as trust or loyalty with several questions (*items*) and average them into a *scale score*. This chapter uses *practice_brand_survey.csv*: 500 simulated respondents who rated Trust1 to Trust4, Value1 to Value4 and Loyalty1 to Loyalty3 on a 1 to 7 scale, with Age_group (18-24, 25-34, 35-49, 50+), Gender and Customer_type (Long-term, New, Regular).

On the **Data** screen, open the file. SaNaML reports 500 rows and 15 variables; Value2 and Loyalty2 each have 18 missing answers. RespondentID is a label and is not used.

3.1 Data quality: straight-lining

Concept: careless responding and straight-lining

Some respondents answer without reading: the same answer everywhere (*straight-lining*), random answers, or far too fast. Their answers weaken correlations and can create or hide effects. No single index catches every kind of careless answering, so the usual advice is to compute several indices and exclude only respondents flagged by more than one (Curran, 2016; Meade & Craig, 2012).

1. On the **Data** screen click **Check data quality** → (step 5). Check the *Screening settings* (Figure 15): all 11 items are ticked, the rule is **Suggest exclusion when flagged by 2 or more indices (Recommended)**, and no completion time is chosen because the file has none.
2. Click **Screen the data**.

Screening settings (What to check)

Questionnaire items (in the order of the file)

Select all Select none

☒ Trust1 ☒ Trust2 ☒ Trust3 ☒ Trust4 ☒ Value1 ☒ Value2 ☒ Value3

☒ Value4 ☒ Loyalty1 ☒ Loyalty2 ☒ Loyalty3

Completion time in seconds — none — Suggest exclusion when flagged by 2 or more indices (Recommended) Longstring flag at half the itx

Reverse-coded items are taken from the construct scores you built; scales come from the constructs, the PLS-SEM model or item names (PE1, PE2 ... form PE).

Screen the data

Figure 15. The screening settings for the brand survey.

Data quality screening (Careless responding, outliers and missing data)

What it means: Careless responding: 3 indices were computed on 11 items. 10 respondents were flagged by 2 or more indices and are suggested for exclusion (2.0% of 500); 7 more raised a single flag, which on its own is not enough to exclude anyone. Outliers: 0 values with $|z| > 3.29$ across 0 variables. Missing data: 0.7% of the answers are missing (at most 3.6% for one item); 464 respondents (93%) answered every item. Little's test does not reject that the data are missing completely at random ($\chi^2(20) = 13.81, p = .840$).

Note Careless responding. 10 respondents (2.0%) are suggested for exclusion. Look at their answers before deciding, and report the rule and the number excluded.

Rule: Use several careless-response indices and exclude only respondents flagged by more than one; report the indices, cut-offs and the number excluded (Curran (2016); Meade & Craig (2012)).

OK Outliers. No value is beyond $|z| = 3.29$.

Rule: Univariate outliers: $|z| > 3.29$ ($p < .001$); multivariate outliers: Mahalanobis D^2 with $p < .001$ (Tabachnick & Fidell (2019)).

OK Amount of missing data. At most 3.6% missing for one item.

Rule: Under 10% missing per variable is usually ignorable if the missingness is random; more needs a principled method (FIML or multiple imputation) (Hair et al. (2019)).

OK Missing completely at random. Little's test: $\chi^2(20) = 13.81, p = .840$. A non-significant result does not prove MCAR, but it gives no reason to doubt it.

Rule: Little's test $p \geq .05$: no evidence against missing completely at random (Little (1988)).

Careless-response indices

Index	Flag when
Longstring (Same answer many times in a row)	6 or more identical answers in a row (of 11 items)
Intra-individual response variability (No variation at all)	SD of the answers = 0
Mahalanobis distance (Unusual answer pattern)	D^2 with $p < .001$ ($\chi^2, 11$ df; complete answers)
Person-total correlation (Answers unlike everyone else's)	Not used: the item means are too similar ($SD\ 0.03 < 0.25$) for the index to be informative
Even-odd consistency (Inconsistent within scales)	Not used: needs at least 4 scales with two or more items (found 3); with fewer scales the
Response time (Too fast to have read the items)	Not used: no completion time chosen

A respondent flagged by at least 2 indices is suggested for exclusion; no single index is decisive. Longstring follows the item order of the file. Reverse-coded items (none) were reversed for the person-total and even-odd indices. Scales: Trust (4), Value (4), Loyalty (3).

Respondents suggested for exclusion

Figure 16. The start of the screening report.

The report (Figure 16) says: "Careless responding: 3 indices were computed on 11 items. 10 respondents were flagged by 2 or more indices and are suggested for exclusion (2.0% of 500); 7 more raised a single flag, which on its own is not enough to exclude anyone." It adds that there are no outliers, that 0.7% of the answers are missing (at most 3.6% for one item), that 464 respondents (93%) answered every item, and that Little's test does not reject that the answers are missing completely at random, $\chi^2(20) = 13.81, p = .840$.

Table 6

The careless-response indices for the brand survey (three are computed, three are not used)

Index	Flag when	Respondents flagged
Longstring (same answer many times in a row)	6 or more identical answers in a row (of 11 items)	17 (3.4%)
Intra-individual response variability (no variation at all)	SD of the answers = 0	10 (2.0%)
Mahalanobis distance (unusual answer pattern)	D^2 with $p < .001$	0 (0.0%)
Person-total correlation	Not used: the item means are too similar ($SD\ 0.03 < 0.25$)	none
Even-odd consistency	Not used: needs at least 4 scales with two or more items (found 3)	none
Response time	Not used: no completion time chosen	none

What it means for our example

Ten respondents (B019, B025, B108, B162, B233, B237, B352, B374, B424 and B496) gave the same answer to all 11 items (longest run 11, standard deviation 0.00): these are the planted straight-liners, and each is flagged by two indices. Seven more respondents (B002, B052, B098, B179, B244, B431 and B461) have a longest run of 6 to 9 identical answers, so only the longstring index flags them; one flag is not enough, and a run of 6 to 9 in an 11-item questionnaire can occur in an honest answer. This is why no single index is enough, and why the rule needs two flags. The even-odd index is switched off for this file, because it is informative only when a questionnaire has at least four scales and this one has three; SaNaML says so in the table instead of reporting an index that would be unreliable.

The card *Your decision (Which respondents to exclude)* lists the 17 respondents with at least one flag; the 10 suggested ones are ticked (Figure 17).

Your decision (Which respondents to exclude)				
Respondents with at least one flag (17); ticked = to be excluded				
Exclude	Respondent	Flags	Longstring	IRV
☐	B002	Longstring	6	0.75
☒	B019	Longstring, Intra-individual response variability	11	0.00
☒	B025	Longstring, Intra-individual response variability	11	0.00
☐	B052	Longstring	6	0.79
☐	B098	Longstring	7	1.19
☒	B108	Longstring, Intra-individual response variability	11	0.00
☒	B162	Longstring, Intra-individual response variability	11	0.00
☐	B179	Longstring	7	0.54
☒	B233	Longstring, Intra-individual response variability	11	0.00
☒	B237	Longstring, Intra-individual response variability	11	0.00
Exclude the 10 ticked respondents				

Figure 17. Your decision: the flagged respondents; the 10 suggested for exclusion are ticked.

1. Look at the flagged respondents and untick anyone you want to keep.
2. Click **Add to Word report** under the screening report, so that the rule, the cut-offs and the number excluded are documented.
3. Click **Exclude the 10 ticked respondents**. The card *Screened data in use* confirms it: "10 of 500 respondents are excluded: B019, B025, B108, B162, B233, B237, B352, B374, B424, B496. Every analysis now uses the remaining 490."

Nothing is deleted from your file, and **Undo the exclusion (use all 500 respondents)** brings everyone back.

How to report it

"Data were screened for careless responding with 3 indices (longstring, intra-individual response variability, Mahalanobis distance; Curran, 2016; Meade & Craig, 2012); 10 of 500 respondents were flagged by at least 2 indices and excluded; 0.7% of item responses were missing, and Little's (1988) MCAR test was not significant, $\chi^2(20) = 13.81$, $p = .840$."

3.2 Reliability and scale scores

Step 3 of the **Data** screen builds a construct (scale) score as the average of its items and first checks that the items belong together.

1. In **Name of the construct** type **Trust**.
2. In **Quick pick: items starting with** type **Trust** and press Tab. Trust1 to Trust4 are ticked, and **Lowest answer** and **Highest answer** are filled in from the data (1 and 7).
3. Read Cronbach's α and the item statistics (Figure 18). No item is reverse-worded here, so leave the **R** boxes empty.
4. Click **Create the construct score**.
5. Repeat for **Value** and **Loyalty**; Table 7 gives the three results.

3 • Build construct scores (optional)

A construct score is the average of the items that measure one idea (for example PE1-PE4 → PE). SaNaML first checks the items hang together (Cronbach's α). If you already have latent variable scores from other software, open that file instead; it needs no construct building.

Name of the construct:

Letters, numbers and underscores; must not be used by another variable yet.

Quick pick: items starting with:

Lowest answer: Highest answer:

Used to reverse-code items: on a 1-7 scale, a reversed 2 becomes 6.

Items (tick to include; "R" = reverse-coded)

☒	Trust1	☐	R
☒	Trust2	☐	R
☒	Trust3	☐	R
☒	Trust4	☐	R
☐	Value1		
☐	Value2		
☐	Value3		
☐	Value4		
☐	Loyalty1		
☐	Loyalty2		

$\alpha = .87$ Cronbach's α on 490 complete responses. Cronbach's α of .70 or higher is usually acceptable for research scales (Nunnally (1978)).

Item	Corrected item-total r	α if deleted
Trust1	.74	.82
Trust2	.70	.83
Trust3	.73	.82
Trust4	.68	.84

An item-total correlation below .30 (or a negative one) suggests the item does not fit, or that it needs reverse-coding.

Create the construct score

Figure 18. Building the Trust score: Cronbach's alpha and item statistics.

Concept: Cronbach's alpha

Cronbach's α summarises how consistently the items of a scale move together. It rises with the number of items and with their average correlation. The corrected item-total correlation is the correlation of an item with the sum of the other items; the column " α if deleted" shows the α the scale would have without that item.

Cut-off values: internal consistency

Cronbach's α of .70 or more is usually acceptable for research scales (Nunnally, 1978). The pill is green at .70 or more, amber from .60 and red below.

Corrected item-total correlation below .30, or negative, suggests that the item does not fit or needs reverse-coding.

Table 7

Reliability of the three scales (490 respondents after screening)

Scale	α (complete cases)	Corrected item-total r	α if item deleted
Trust (4 items)	.87 (n = 490)	.74, .70, .73, .68	.82, .83, .82, .84
Value (4 items)	.80 (n = 472)	.62, .65, .61, .61	.76, .74, .76, .76
Loyalty (3 items)	.84 (n = 472)	.74, .69, .69	.75, .79, .79

What it means for our example

All three scales are reliable (α between .80 and .87), and deleting an item would lower α every time, so all items stay. α is computed only for respondents who answered all items of the scale: Value2 and Loyalty2 each have 18 missing answers, which is why the Value and Loyalty scores exist for 472 respondents and the Trust score for all 490. If your questionnaire has reverse-worded items, tick **R** for them: otherwise they correlate negatively with the others.

Tip: the construct score is the average of its items

A respondent who skipped any item of a scale gets no score for it. Check the **Valid** and **Missing** columns in step 2 after creating the scores.

Describe the scores with **Analyses** → **Descriptive statistics** (Figure 19).

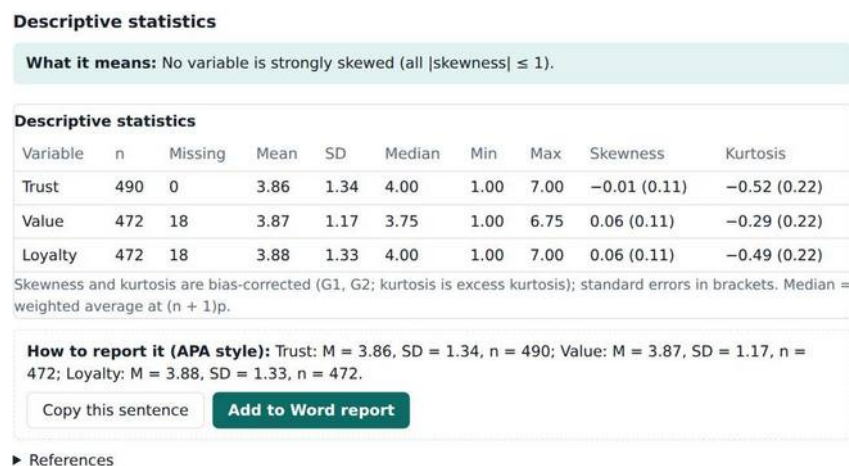


Figure 19. Descriptive statistics of the three scale scores.

Table 8

Descriptive statistics of the scale scores (1 to 7 scale)

Score	n	Mean	SD	Median	Min	Max	Skewness
Trust	490	3.86	1.34	4.00	1.00	7.00	-0.01
Value	472	3.87	1.17	3.75	1.00	6.75	0.06
Loyalty	472	3.88	1.33	4.00	1.00	7.00	0.06

No variable is strongly skewed (all $|\text{skewness}| \leq 1$).

How to report it

"Trust (4 items, $\alpha = .87$), Value (4 items, $\alpha = .80$) and Loyalty (3 items, $\alpha = .84$) were computed as the mean of their items on a 1–7 scale."

3.3 Compare loyalty between groups

Choose **Analyses** → **One-way ANOVA**, **Loyalty** as the **Score to compare** and **Customer_type** as the **Groups** (Figure 20).

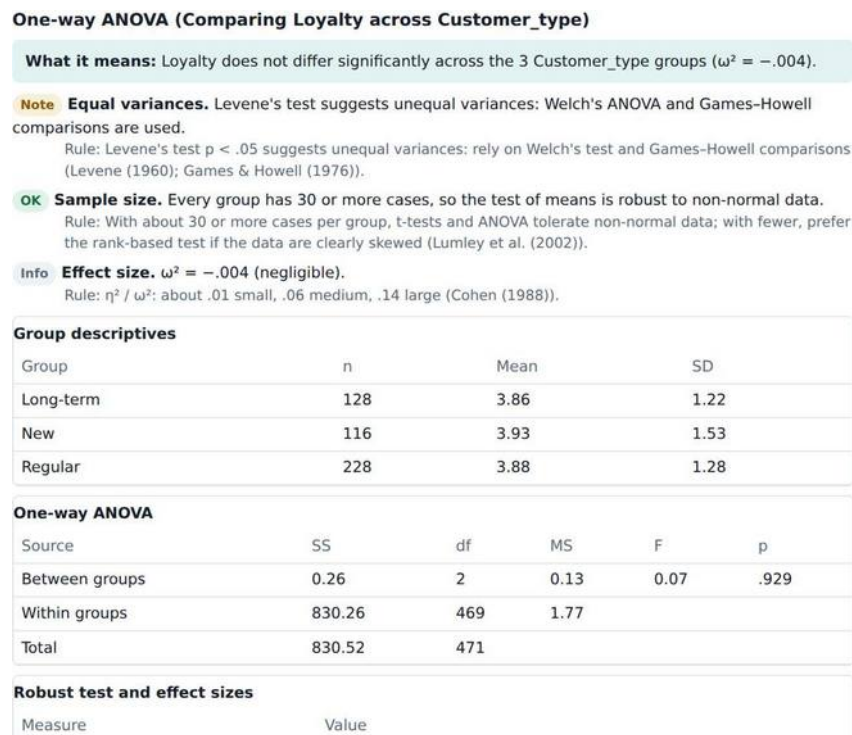


Figure 20. One-way ANOVA of Loyalty by Customer_type.

SaNaML reports the classic ANOVA, Welch's F (robust to unequal variances), the effect sizes η^2 and ω^2 , pairwise comparisons and the rank-based Kruskal-Wallis test.

What it means for our example

Mean loyalty is almost the same for long-term (3.86, $n = 128$), new (3.93, $n = 116$) and regular customers (3.88, $n = 228$). Levene's test suggests unequal variances ($F(2, 469) = 4.33$, $p = .014$; new customers vary most, $SD = 1.53$), so SaNaML uses Welch's test: $F(2, 251.20) = 0.06$, $p = .940$, $\omega^2 = -.004$. The Kruskal-Wallis test agrees, $H(2) = 0.17$, $p = .919$. Customer type does not matter for loyalty in this sample.

Now compare Loyalty by **Age_group** (Figure 21).

One-way ANOVA (Comparing Loyalty across Age_group)

What it means: Loyalty does not differ significantly across the 4 Age_group groups ($\omega^2 = .005$).

OK Equal variances. Levene's test does not suggest unequal variances: the classic ANOVA and Tukey-Kramer comparisons are used.

Rule: Levene's test $p < .05$ suggests unequal variances: rely on Welch's test and Games-Howell comparisons (Levene (1960); Games & Howell (1976)).

OK Sample size. Every group has 30 or more cases, so the test of means is robust to non-normal data.

Rule: With about 30 or more cases per group, t-tests and ANOVA tolerate non-normal data; with fewer, prefer the rank-based test if the data are clearly skewed (Lumley et al. (2002)).

Info Effect size. $\omega^2 = .005$ (negligible).

Rule: η^2 / ω^2 : about .01 small, .06 medium, .14 large (Cohen (1988)).

Group descriptives				
Group	n	Mean	SD	
18-24	96	3.88	1.33	
25-34	128	3.68	1.30	
35-49	144	4.05	1.31	
50+	104	3.91	1.37	

One-way ANOVA					
Source	SS	df	MS	F	p
Between groups	9.07	3	3.02	1.72	.161
Within groups	821.45	468	1.76		
Total	830.52	471			

Robust test and effect sizes					
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Figure 21. One-way ANOVA of Loyalty by Age_group.

What it means for our example

The four age groups differ in the sample (18-24: 3.88; 25-34: 3.68; 35-49: 4.05; 50+: 3.91), but not significantly: $F(3, 468) = 1.72$, $p = .161$, $\eta^2 = .011$, $\omega^2 = .005$; Kruskal-Wallis $H(3) = 4.55$, $p = .208$. The largest gap, between 25-34 and 35-49 (0.36 points), has an adjusted p of .109. Do not report this as an age effect.

How to report it

"Loyalty did not differ between age groups, $F(3, 468) = 1.72$, $p = .161$, $\eta^2 = .011$."

3.4 What drives loyalty: correlation and regression

Choose **Analyses** → **Correlation and regression analysis** → **Bivariate and partial correlation**, and tick Trust, Value and Loyalty (Figure 22).

Pearson product-moment correlation (Pearson r)

What it means: 3 of 3 correlations are significant after Holm adjustment. The strongest is Trust with Loyalty ($r = .63$, large, higher goes with higher).

Info Strength. Read the size of a correlation from its absolute value.

Rule: Correlation-type effect sizes (r , rank-biserial, ϕ): about .10 small, .30 medium, .50 large (Cohen (1988)).

Info Method. Pearson's r measures straight-line relationships. With strongly skewed data or many tied ratings, Spearman's ρ is a safer choice.

Correlation matrix

	1	2	3
1. Trust	—		
2. Value	.37***	—	
3. Loyalty	.63***	.46***	—

* $p < .05$, ** $p < .01$, *** $p < .001$ (Holm-adjusted). Each pair uses every respondent with both values.

Every pair

Pair	r	95% CI	n	p	p (Holm)
Trust - Value	.369	[.29, .44]	472	< .001	< .001
Trust - Loyalty	.625	[.57, .68]	472	< .001	< .001
Value - Loyalty	.460	[.38, .53]	454	< .001	< .001

How to report it (APA style): Pearson r correlations (Holm-adjusted p -values) showed: Trust and Loyalty, $r(470) = .63$, $p < .001$, 95% CI [.57, .68]; Value and Loyalty, $r(452) = .46$, $p < .001$, 95% CI [.38, .53]; Trust and Value, $r(470) = .37$, $p < .001$, 95% CI [.29, .44].

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► References

Figure 22. Pearson correlations of the three scale scores.

Trust and Loyalty correlate $r = .63$ (95% CI [.57, .68]), Value and Loyalty $r = .46$ ([.38, .53]) and Trust and Value $r = .37$ ([.29, .44]); all $p < .001$ after the Holm adjustment for several tests.

Now regress Loyalty on Trust and Value: choose **Multiple linear regression (Which variables predict a score?)**, **Loyalty** as the **Outcome (scale)**, and tick Trust and Value as **Block 1 predictors** (Figure 23).

Multiple linear regression (Predicting Loyalty)

What it means: The predictors together explain 45.4% of the variance in Loyalty ($R^2 = .45$, $F^2 = 0.83$, large). Significant predictors: Trust ($\beta = .53$), Value ($\beta = .26$).

OK Sample size. 454 respondents for 2 predictor terms (needed: 66).

Rule: At least $50 + 8k$ respondents for k predictors to test the whole model (Green (1991)).

OK Collinearity. Largest VIF = 1.16.

Rule: VIF below 5: no concern; 5-10: moderate; above 10: serious collinearity (Hair et al. (2019); Kutner et al. (2005)).

OK Normal residuals. Shapiro-Wilk $W = 0.997$, $p = .514$.

Rule: Shapiro-Wilk $p \geq .05$ for the residuals; with about 300 or more respondents the coefficient tests are robust anyway (Shapiro & Wilk (1965)).

OK Constant error variance. Breusch-Pagan (Koenker) $\chi^2(2) = 0.27$, $p = .873$.

Rule: Breusch-Pagan (Koenker) $p \geq .05$: no evidence that the error spread changes (Breusch & Pagan (1979); Koenker (1981)).

OK Influential cases. Largest Cook's distance = 0.025; 1 standardised residual beyond ± 3 .

Rule: No Cook's distance above 1 (Cook & Weisberg (1982)).

Info Independent errors. Durbin-Watson = 2.10 (values near 2 mean no order effect; it matters only when the row order means something, such as time).

Info Effect size. $F^2 = 0.83$ (large).

Rule: F^2 : .02 small, .15 medium, .35 large (Cohen (1988)).

Model summary

Model	Predictors	R^2	Adjusted R^2	F	df	p	ΔR^2	p (ΔR^2)
1	Trust, Value	.454	.452	187.86	2, 451	< .001		

Coefficients (model 1)

Predictor	B	SE	β	t	p	95% CI of B	VIF
(Intercept)	0.717	0.178		4.03	< .001	[0.367, 1.066]	

Figure 23. The regression result: what it means, the checks and the model summary.

What it means for our example

Trust and Value together explain 45.4% of the variance in Loyalty ($R^2 = .45$, adjusted $R^2 = .452$; $F(2, 451) = 187.86$, $p < .001$). Both predict loyalty: Trust ($B = 0.519$, $\beta = .53$, $t(451) = 14.16$, $p < .001$) more strongly than Value ($B = 0.297$, $\beta = .26$, $t(451) = 7.06$, $p < .001$). One more point of trust goes with about half a point more loyalty when value is held constant. All checks are fine: the largest VIF is 1.16, residuals are normal (Shapiro–Wilk $p = .514$), the error spread is constant (Breusch–Pagan $p = .873$) and no case is influential (largest Cook's distance 0.025). The regression used 454 respondents; 36 with a missing item were left out.

Cut-off values: regression

Sample size: at least $50 + 8k$ respondents for k predictors (Green, 1991). **VIF:** below 5 no concern (Hair et al., 2019). **f^2 :** .02 small, .15 medium, .35 large (Cohen, 1988).

Note: correlation is not causation

The data are a single survey. A regression shows which scores go together with loyalty, not that raising trust would raise loyalty. For marketing decisions the direction is plausible but needs an experiment or longitudinal data.

How to report it

"A multiple linear regression of Loyalty on Trust and Value was significant, $F(2, 451) = 187.86$, $p < .001$, $R^2 = .45$. Trust predicted Loyalty, $\beta = .53$, $t(451) = 14.16$, $p < .001$; Value predicted Loyalty, $\beta = .26$, $t(451) = 7.06$, $p < .001$."

Exercises

Exercise: survey scales

1. Run the screening with **Suggest exclusion when flagged by** set to **any index (Strict)**. How many respondents are suggested now? Compare with the 10 of the recommended rule (the card *Your decision* lists 17 respondents with at least one flag), and decide which rule you would defend in a report.

1. Compare Trust by Gender with an independent-samples t-test. The practice guide lists no gender effect; report what you find, with its effect size.

1. Planted answer for the chapter: Loyalty depends on both Trust and Value (section 3.4), Customer_type has no effect (section 3.3), and ten respondents straight-lined (section 3.1).

4. Testing a campaign

Practice_marketing.csv contains a randomised campaign test: each customer was randomly given either the *Standard* offer (762 customers) or the *New offer* (738 customers), and the file records whether the customer Responded, and the monthly spend before and after the campaign. Load the file again (Data → Choose a file...).

4.1 Did more customers respond? The A/B test of a rate

Concept: A/B test of a response rate

An A/B test compares two versions of an offer given to comparable customers. Because the groups were formed at random, a difference in the response rate can be attributed to the offer rather than to who received it. The chi-square test of independence asks whether the response rate differs between the groups by more than chance would produce.

1. Choose **Analyses** → **Chi-square, Fisher's exact and McNemar tests** and **Chi-square test of independence (Are two categorical variables related?)**.
2. Set the first variable (**Categorical variable**) to **Campaign_group** and the **Second categorical variable** to **Responded**.
3. Read the result (Figure 24).

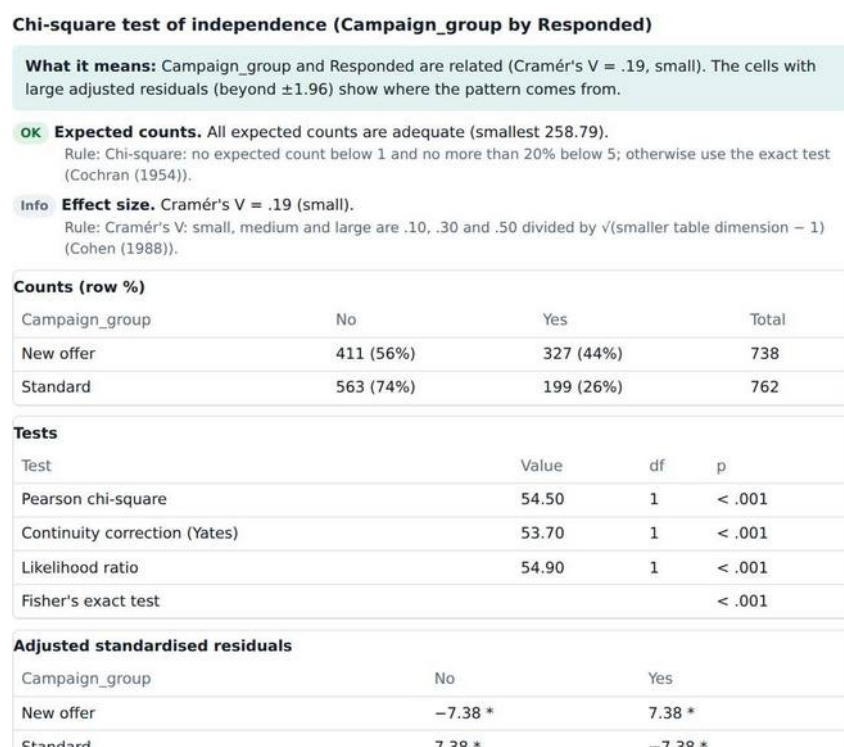


Figure 24. The chi-square test of Responded by Campaign_group.

Table 9

Responded by Campaign_group (counts and row percentages)

Group	No	Yes	Total
New offer	411 (56%)	327 (44%)	738

Group	No	Yes	Total
Standard	563 (74%)	199 (26%)	762

What it means for our example

44% of the customers who received the New offer responded, against 26% of those who received the Standard offer: a difference of about 18 percentage points (327 of 738 against 199 of 762). The difference is statistically significant, $\chi^2(1, N = 1500) = 54.50, p < .001$. The effect size is Cramér's $V = .19$, which SaNaML labels *small*; the adjusted residuals (± 7.38) show that the pattern comes from both cells: more responders than expected after the New offer, fewer after the Standard offer. All expected counts are adequate (smallest 258.79), so Fisher's exact test ($p < .001$) and the continuity-corrected test ($\chi^2 = 53.70$) agree.

Concept: statistical significance and practical importance

Cohen's labels (V of .10 small, .30 medium, .50 large for a 2×2 table) describe the size of an association in general, not its commercial value. A "small" V can correspond to a large business difference: here, 18 more responders per 100 customers contacted. Always report the two rates, not only the p -value.

How to report it

"A chi-square test of independence showed a significant association between campaign group and response, $\chi^2(1, N = 1500) = 54.50, p < .001$, Cramér's $V = .19$: 44% of customers who received the new offer responded, compared with 26% who received the standard offer."

4.2 Did monthly spend change? The paired test

Each customer has Monthly_spend_before and Monthly_spend_after. Choose **Analyses** → **Independent-samples, paired-samples and one-sample t-tests** → **Paired-samples t-test (The same people measured twice)**, and choose Monthly_spend_before as the first and Monthly_spend_after as the second measurement (Figure 25).

Paired-samples t-test (Comparing Monthly_spend_before and Monthly_spend_after, same respondents)

What it means: Monthly_spend_after is higher than Monthly_spend_before by 21.71 points on average. The change is statistically significant and medium ($d_z = 0.74$).

OK Sample size. Every group has 30 or more cases, so the test of means is robust to non-normal data.
Rule: With about 30 or more cases per group, t-tests and ANOVA tolerate non-normal data; with fewer, prefer the rank-based test if the data are clearly skewed (Lumley et al. (2002)).

Info Effect size. $d_z = 0.74$ is medium (mean change divided by the SD of the changes).
Rule: Cohen's d: about 0.2 small, 0.5 medium, 0.8 large (Cohen (1988)).

Descriptives (complete pairs)						
Measure	n	Mean	SD	Median		
Monthly_spend_before	1500	282.64	198.64	238.45		
Monthly_spend_after	1500	304.35	215.75	255.80		
Difference (Monthly_spend_after – Monthly_spend_before)	1500	21.71	29.22	13.50		

Paired t-test (two-sided)						
t	df	p	Mean difference	95% CI	d_z	CI of d_z
28.78	1499	< .001	21.71	[20.23, 23.19]	0.74	[0.69, 0.80]

Rank-based alternative						
Test	V	z	p	Matched rank-biserial r	Method	
Wilcoxon signed-rank	1010454.0	30.61	< .001	.93	normal approximation	

53 pairs with no change were left out of the signed-rank test.

How to report it (APA style): A paired-samples t-test showed that Monthly_spend_after ($M = 304.35$, $SD = 215.75$) differed from Monthly_spend_before ($M = 282.64$, $SD = 198.64$), $t(1499) = 28.78$, $p < .001$, $d_z = 0.74$, 95% CI [0.69, 0.80].

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► References

Figure 25. The paired-samples t-test of monthly spend before and after the campaign.

Table 10

Monthly spend before and after the campaign ($n = 1500$)

Measure	Mean	SD	Median
Monthly_spend_before	282.64	198.64	238.45
Monthly_spend_after	304.35	215.75	255.80
Difference (after – before)	21.71	29.22	13.50

What it means for our example

Spend rose by 21.71 per month on average, $t(1499) = 28.78$, $p < .001$, 95% CI of the mean difference [20.23, 23.19], $d_z = 0.74$ (95% CI [0.69, 0.80]), a medium standardised change. The rank-based alternative agrees: the Wilcoxon signed-rank test gives $z = 30.61$, $p < .001$, matched rank-biserial $r = .93$ (53 customers with no change were left out of this test). The mean difference (21.71) is larger than the median difference (13.50), because some customers rose by much more than others.

Note: a before-after change is not an effect of the campaign

The paired test compares everyone's spend before and after, whichever offer they received. Spend may have risen for other reasons (season, prices, other promotions). It cannot tell you whether the *offer* mattered. To judge the offer, compare the *change* between the two campaign groups (section 4.3).

4.3 Did the New offer raise spend more than the Standard offer?

A change score is the difference between two variables of the same customer, and step 4 of the **Data** screen makes it (Figure 26).

1. On the **Data** screen, in step 4 · *Compute and recode variables (optional)*, choose **Combine two scale variables (difference, change, ratio ...)**.
2. Choose **Monthly_spend_after** as **Variable A**, keep **Difference, A – B**, and choose **Monthly_spend_before** as **Variable B**.
3. Type the name **Spend_change** in **Name of the new variable** and click **Add the new variable**. SaNaML confirms: "Added Spend_change: 1500 of 1500 respondents have a value."

4 · Compute and recode variables (optional)

Make a new variable from an existing one: a logarithm for a skewed variable, a standardised score, low and high groups, a 0/1 indicator for one category, or merged categories. The original variable is never changed.

What do you want to make?
Combine two scale variables (difference, change, ratio ...)

Variable A
Monthly_spend_after

How to combine A (the variable above) and B
Difference, A – B

Variable B
Monthly_spend_before

A change or gain score: after minus before.

Name of the new variable
Spend_change
Letters, numbers, dots and underscores; must not be used by another variable yet.

Add the new variable

Figure 26. Making $Spend_change = Monthly_spend_after - Monthly_spend_before$.

Then choose an independent-samples t-test with **Score to compare** = **Spend_change** and **Groups** = **Campaign_group** (Figure 27).

Independent-samples t-test (Comparing Spend_change between New offer and Standard)

What it means: New offer scored higher than Standard on Spend_change (difference 20.52 points). The difference is statistically significant and its size is medium ($|d| = 0.75$).

Info Which t-test. Welch's t-test is reported first because it is accurate whether or not the variances are equal.

Rule: Compare two means with Welch's t-test by default; it stays accurate when the variances differ (Delacre et al. (2017)).

Note Equal variances. Levene's test suggests the variances differ; rely on Welch's test.

Rule: Levene's test $p < .05$ suggests unequal variances: rely on Welch's test and Games-Howell comparisons (Levene (1960); Games & Howell (1976)).

OK Sample size. Every group has 30 or more cases, so the test of means is robust to non-normal data.

Rule: With about 30 or more cases per group, t-tests and ANOVA tolerate non-normal data; with fewer, prefer the rank-based test if the data are clearly skewed (Lumley et al. (2002)).

Info Effect size. $|d| = 0.75$ is medium. The sign of d follows the order of the groups: $d = 0.75$ means New offer scored higher than Standard.

Rule: Cohen's d: about 0.2 small, 0.5 medium, 0.8 large (Cohen (1988)).

Group descriptives					
Group	n	Mean	SD	Median	Skewness
New offer	738	32.13	37.34	18.30	1.83
Standard	762	11.61	11.12	10.70	0.37

t-tests (two-sided)					
Test	t	df	p	Mean difference	95% CI
Welch's t (recommended)	14.33	863.00	< .001	20.52	[17.71, 23.33]
Student's t (equal variances)	14.52	1498	< .001	20.52	[17.75, 23.29]

Figure 27. The change in monthly spend, New offer against Standard.

Table 11

Change in monthly spend by campaign group

Group	n	Mean change	SD	Median	Skewness
New offer	738	32.13	37.34	18.30	1.83
Standard	762	11.61	11.12	10.70	0.37

What it means for our example

Spend rose by 32.13 per month after the New offer and by 11.61 after the Standard offer. Welch's t-test shows that the difference of 20.52 is statistically significant, $t(863.00) = 14.33$, $p < .001$, $d = 0.75$, 95% CI [0.65, 0.85]. SaNaML's effect-size line reads " $|d| = 0.75$ is medium. The sign of d follows the order of the groups: $d = 0.75$ means New offer scored higher than Standard." Levene's test shows that the variances differ strongly ($F(1, 1498) = 409.54$, $p < .001$): the New offer produced a few very large increases (skewness 1.83), so the mean change (32.13) lies well above the median (18.30). SaNaML therefore reports Welch's t-test first, and the rank-based Mann-Whitney test agrees ($z = 11.07$, $p < .001$, $r = .33$). Both the response rate and the spend change favour the New offer.

The same step can express the change in per cent: choose **Percentage change, $(A - B) \div B \times 100$** with the same two variables and the name **Spend_change_pct**. SaNaML reports "Added Spend_change_pct: 1418 of 1500 respondents have a value", because the 82 customers whose spend before was 0 have no percentage. The independent-samples t-test of Spend_change_pct by Campaign_group agrees with the test of the difference: the mean increase is 10.91% after the New

offer (SD = 9.79, n = 696) and 3.86% after the Standard offer (SD = 5.12, n = 722), Welch's $t(1040.41) = 16.90$, $p < .001$, $|d| = 0.91$, 95% CI [0.80, 1.02].

Tip: check that randomisation produced comparable groups

Compare a variable measured before the test, or one that the offer cannot have changed, between the groups. The annual spend does not differ between the campaign groups: Welch's $t(1492.71) = -0.18$, $p = .859$, $d = -0.01$ (means 3380.61 and 3402.45). SaNaML explains the sign: $d = -0.01$ means that Standard scored higher than New offer, by a negligible amount ($|d| = 0.01$). This is what random assignment should give.

How to report it

"Monthly spend increased more after the new offer (M change = 32.13, SD = 37.34) than after the standard offer (M change = 11.61, SD = 11.12), Welch's $t(863.00) = 14.33$, $p < .001$, $d = 0.75$, 95% CI [0.65, 0.85]."

4.4 How many customers does a campaign test need?

Power analysis answers the question before the test is run: how many customers are needed to detect a difference of a given size with a given probability? No data are needed. Click **Power analysis**.

1. In 1 · Which analysis will you run? choose **Chi-square test (Goodness of fit or independence)**.
2. In 2 · What do you want to find? choose **A priori power analysis (The sample size I need)**.
3. Enter the effect size Cohen's w , $\alpha = .05$, power = .80 and the degrees of freedom (1 for a 2×2 table).

1 - Which analysis will you run?

- ☐ **Multiple linear regression, F-test of R² (also the usual PLS-SEM check)**
For PLS-SEM, use the largest number of arrows pointing at one construct as the number of predictors.
- ☐ **Inverse square root method (PLS-SEM minimum sample size)**
Minimum sample size from the smallest path coefficient you expect (Kock & Hadaya, 2018).
- ☐ **Independent-samples t-test (Compare two groups)**
Two groups of different people.
- ☐ **Paired-samples t-test (Same people twice)**
Before vs after, or two ratings per person.
- ☐ **One-sample t-test (One score against a fixed value)**
Compares one mean with a fixed value.
- ☐ **One-way ANOVA, fixed effects (Compare three or more groups)**
One-way ANOVA, effect size f.
- ☒ **Chi-square test (Goodness of fit or independence)**
Goodness of fit or independence; effect size w.
- ☐ **Pearson correlation test (Is a correlation different from zero?)**
Test whether a correlation differs from zero.

2 - What do you want to find?

- ☒ **A priori power analysis (The sample size I need)**
Most common in proposals.
- ☐ **Post hoc power analysis (The power of a given sample size)**
For a fixed number of respondents.
- ☐ **Sensitivity power analysis (The smallest effect I can detect)**
Useful when the sample size is already fixed.

Effect size Cohen's w **Significance level α** **Power (1 - β)** **Degrees of freedom**

0.19 **0.05** **0.8** **1**

Conventions: small 0.1, medium 0.3, large 0.5 (Cohen, 1988). Better: take it from earlier studies.

Plan for a power of .80 at $\alpha = .05$ unless the field expects otherwise.

Goodness of fit: categories - 1. Independence: (rows - 1) $\times$ (columns - 1).

N = 218

How to report it: An a priori power analysis for a chi-square test with 1 df indicated that 218 participants are needed to detect Cohen's w = 0.19 with $\alpha = .05$ and power = .80 (achieved power .801).

► References

Figure 28. A priori power analysis for a chi-square test with one degree of freedom ($w = 0.19$).

Concept: effect size w, and where to get it

For a 2×2 table, Cohen's w equals the phi coefficient, which is Cramér's V for a table with one degree of freedom. The A/B test above had $V = .19$, so $w = 0.19$ is the observed effect. Take the planning value from a pilot test, from an earlier campaign or from the smallest difference that would matter commercially. Conventions (small 0.1, medium 0.3, large 0.5; Cohen, 1988) are a last resort.

Table 12

Sample sizes for campaign tests ($\alpha = .05$, power = .80, two-sided)

Test	Effect size	Total sample size
Chi-square, 1 df	$w = 0.19$ (the observed A/B effect)	218 customers
Chi-square, 1 df	$w = 0.10$ (a small effect)	785 customers
Independent-samples t-test	$d = 0.20$ (small)	788 customers (394 per group)
Paired-samples t-test	$d_z = 0.20$ (small)	199 customers

What it means for our example

To detect an effect of the size we observed ($w = 0.19$) with a power of .80, about 218 customers in total are enough: the test with 1500 customers was ample. To detect a small difference in response rates ($w = 0.10$), 785 customers are needed. SaNaML writes the sentence for your plan: "An a priori power analysis for a chi-square test with 1 df indicated that 218 participants are needed to detect Cohen's $w = 0.19$ with $\alpha = .05$ and power = .80 (achieved power .801)."

Choose **Sensitivity power analysis (The smallest effect I can detect)** when the sample size is fixed. With $N = 300$ customers and power .80, a chi-square test with one degree of freedom can detect effects of $w = 0.162$ or larger.

Note: power for tests that are not on the list

The power screen covers t-tests, one-way ANOVA, regression, chi-square and correlation tests. A comparison of two proportions can be planned with the chi-square option as shown above. Always plan for the smallest effect worth acting on, and for missing responses.

Exercises

Exercise: the campaign

1. Repeat the A/B test for Responded by Loyalty_member. Do members respond more often? (Compare with the logistic regression in section 6.2, where loyalty members have 2.51 times the odds of responding.)
 1. With the sensitivity analysis, find the smallest w detectable with $N = 1500$ and power .80, and compare it with the observed $V = .19$.
 1. Planted answer for the chapter: New offer response 44% against 26% for Standard ($\chi^2(1) = 54.50$), and a spend rise after the campaign (paired $t(1499) = 28.78$) that is larger after the New offer (32.13 against 11.61).

5. What drives customer spend

Which customer characteristics go with higher annual spend, and do groups of customers spend differently? We use `practice_marketing.csv` and, for section 5.2, the two logarithm variables made in section 2.6 (add them again with step 4 of the **Data** screen if you have reopened the file).

5.1 Multiple linear regression

Concept: multiple regression

A multiple regression predicts a score (here `Annual_spend`) from several predictors at once. Each coefficient B is the change in the outcome for one more unit of that predictor *with the other predictors held constant*. The standardised coefficient β puts predictors on a common scale, and R^2 is the share of the variance of the outcome that the model explains.

Choose **Analyses** → **Correlation and regression analysis** → **Multiple linear regression (Which variables predict a score?)**, choose **Annual_spend** as the **Outcome (scale)** and tick as **Block 1 predictors** `Income_k`, `Tenure_months`, `Loyalty_member`, `Visits_per_month` and `Satisfaction_1to10`. Categorical predictors, marked "(groups)", are dummy-coded against their first group, so `Loyalty_member` appears as "Loyalty_member: Yes (vs No)".

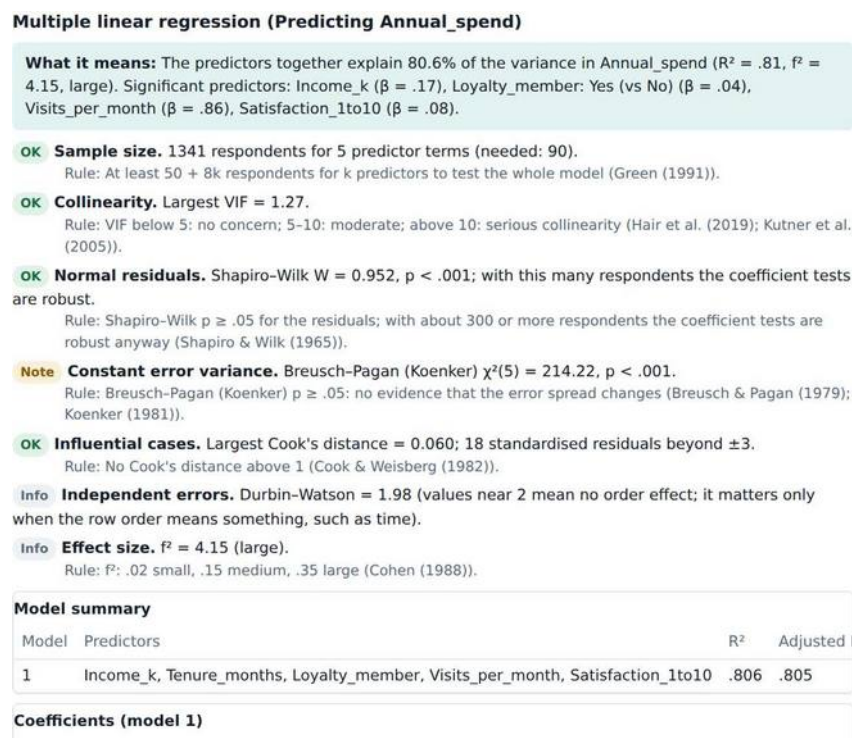


Figure 29. The regression of `Annual_spend`: what it means, the checks and the model summary.

Table 13

Regression of `Annual_spend` ($n = 1341$; 159 customers with a missing value left out)

Predictor	B	β	t	p
Income_k	16.071	.169	14.03	< .001
Tenure_months	-0.456	-.004	-0.32	.747

Predictor	B	β	t	p
Loyalty_member: Yes (vs No)	217.219	.045	3.29	.001
Visits_per_month	1057.889	.860	66.81	< .001
Satisfaction_1to10	114.695	.075	5.84	< .001

What it means for our example

The five predictors explain 80.6% of the variance of Annual_spend ($R^2 = .81$, adjusted $R^2 = .805$; $F(5, 1335) = 1108.12$, $p < .001$). Visits per month dominate ($\beta = .86$: each extra monthly visit goes with about 1058 more spend per year). Income ($\beta = .17$), satisfaction ($\beta = .08$) and loyalty membership ($\beta = .04$) add a little; tenure adds nothing. Collinearity is not a problem (largest VIF = 1.27). One check needs attention: the Breusch-Pagan test ($\chi^2(5) = 214.22$, $p < .001$) shows that the spread of the errors is not constant, as expected for a skewed money variable, and 18 standardised residuals lie beyond ± 3 .

Cut-off values: regression checks

Normal residuals: Shapiro-Wilk $p \geq .05$; with about 300 or more customers the coefficient tests are robust anyway (Shapiro & Wilk, 1965). **Constant error variance:** Breusch-Pagan (Koenker) $p \geq .05$ (Breusch & Pagan, 1979; Koenker, 1981). **Influential cases:** no Cook's distance above 1 (Cook & Weisberg, 1982).

5.2 The same model on the logarithm

Choose **Annual_spend_In1p** and then **Annual_spend_In** as the outcome with the same predictors (Figure 30 shows the second).

Multiple linear regression (Predicting Annual_spend_In)

What it means: The predictors together explain 77.3% of the variance in Annual_spend_In ($R^2 = .77$, $F = 3.40$, large). Significant predictors: Income_k ($\beta = .18$), Loyalty_member: Yes (vs No) ($\beta = .05$), Visits_per_month ($\beta = .84$), Satisfaction_1to10 ($\beta = .06$).

OK Sample size. 1268 respondents for 5 predictor terms (needed: 90).

Rule: At least 50 + 8k respondents for k predictors to test the whole model (Green (1991)).

OK Collinearity. Largest VIF = 1.25.

Rule: VIF below 5: no concern; 5–10: moderate; above 10: serious collinearity (Hair et al. (2019); Kutner et al. (2005)).

OK Normal residuals. Shapiro-Wilk $W = 0.997$, $p = .013$; with this many respondents the coefficient tests are robust.

Rule: Shapiro-Wilk $p \geq .05$ for the residuals; with about 300 or more respondents the coefficient tests are robust anyway (Shapiro & Wilk (1965)).

OK Constant error variance. Breusch-Pagan (Koenker) $\chi^2(5) = 2.97$, $p = .705$.

Rule: Breusch-Pagan (Koenker) $p \geq .05$: no evidence that the error spread changes (Breusch & Pagan (1979); Koenker (1981)).

OK Influential cases. Largest Cook's distance = 0.021; 0 standardised residuals beyond ± 3 .

Rule: No Cook's distance above 1 (Cook & Weisberg (1982)).

Info Independent errors. Durbin-Watson = 2.02 (values near 2 mean no order effect; it matters only when the row order means something, such as time).

Info Effect size. $F^2 = 3.40$ (large).

Rule: F^2 : .02 small, .15 medium, .35 large (Cohen (1988)).

Model summary

Model	Predictors	R^2	Adjusted R^2
1	Income_k, Tenure_months, Loyalty_member, Visits_per_month, Satisfaction_1to10	.773	.772

Coefficients (model 1)

Figure 30. The regression of Annual_spend_In (customers with zero spend left out).

Table 14

The same five predictors for three versions of the outcome

Outcome	n	R^2	Residual spread (Breusch-Pagan)	Residuals beyond ± 3
Annual_spend	1341	.806	$\chi^2(5) = 214.22$, $p < .001$	18
Annual_spend_In1p (zeros kept)	1341	.415	$\chi^2(5) = 194.05$, $p < .001$	73
Annual_spend_In (zeros left out)	1268	.773	$\chi^2(5) = 2.97$, $p = .705$	0

What it means for our example

Keeping the zeros (Annual_spend_In1p) makes the model much worse ($R^2 = .415$; only Visits_per_month is significant, $\beta = .64$), because customers with zero spend are extreme outliers on the logarithmic scale (73 residuals beyond ± 3 , Shapiro-Wilk $W = 0.630$). Leaving them out (Annual_spend_In) gives a well-behaved model: constant error spread ($p = .705$), no extreme residuals and $R^2 = .773$. Its coefficients can be read as percentage changes: each extra monthly visit multiplies spend by about 1.36 ($B = 0.309$, $e^{0.309} \approx 1.36$, that is 36% more), loyalty membership by about 1.08 ($B = 0.074$), each extra thousand of income by about 1.005 ($B = 0.005$), and each satisfaction point by about 1.03 ($B = 0.026$); tenure has no effect ($B = 0.000$, $p = .890$).

Note: choose the model that answers your question

The raw model fits well on the whole sample but violates the equal-spread assumption. The logarithmic model on buyers meets the assumptions and describes *how much* buyers spend; it ignores the 82 customers who spent nothing, who need a separate question (who buys at all?), for example with a logistic regression. The guide to the practice data suggests that the logarithm gives R^2 of about .4: this is the Annual_spend_ln1p result ($R^2 = .415$, against .806 for the raw outcome), and it is low only because of the zero-spend customers.

5.3 Visits and basket are parts of spend, not drivers

In the raw model, visits explain most of the variance. Annual spend is built from visits and basket value, so it is no surprise. Try a hierarchical regression: choose **Annual_spend** as the outcome, put Income_k, Tenure_months, Loyalty_member and Satisfaction_1to10 in **Block 1 predictors** and Visits_per_month and Avg_basket in **Block 2 predictors**.

Table 15

Two-block regression of Annual_spend (n = 1341)

Model	R^2	Adjusted R^2	F (df)	ΔR^2
1: Income_k, Tenure_months, Loyalty_member, Satisfaction_1to10	.157	.154	62.05 (4, 1336)	
2: add Visits_per_month, Avg_basket	.890	.890	1802.10 (6, 1334)	.734

What it means for our example

Customer characteristics alone explain 15.7% of the variance of annual spend. Adding visits and basket value raises R^2 by .734 ($p < .001$) to .890. In the full model, Visits_per_month ($\beta = .85$, $t(1334) = 87.61$) and Avg_basket ($\beta = .36$, $t(1334) = 32.01$) are the only significant predictors; income ($B = -0.806$, $p = .425$), loyalty membership ($B = -57.331$, $p = .255$) and satisfaction ($p = .924$) no longer matter.

Note: do not adjust away the mechanism

Loyalty members visit more (section 2.1). When visits are in the model, the coefficient of loyalty shows only what loyalty adds *beyond* the extra visits, which is why it shrinks. Whether loyalty *causes* the extra visits cannot be decided from this table. For marketing questions ask first which variables are outcomes of the programme and which are background, and put only background variables in the model when you want the total difference between members and non-members.

5.4 Compare spend between two groups

Choose **Analyses** → **t-tests** → **Independent-samples t-test**, **Annual_spend** as the **Score to compare** and **Loyalty_member** as the **Groups** (Figure 31).

Independent-samples t-test (Comparing Annual_spend between No and Yes)

What it means: Yes scored higher than No on Annual_spend (difference 1752.36 points). The difference is statistically significant and its size is medium ($|d| = 0.79$).

Info Which t-test. Welch's t-test is reported first because it is accurate whether or not the variances are equal.

Rule: Compare two means with Welch's t-test by default; it stays accurate when the variances differ (Delacre et al. (2017)).

Note Equal variances. Levene's test suggests the variances differ; rely on Welch's test.

Rule: Levene's test $p < .05$ suggests unequal variances; rely on Welch's test and Games-Howell comparisons (Levene (1960); Games & Howell (1976)).

OK Sample size. Every group has 30 or more cases, so the test of means is robust to non-normal data.

Rule: With about 30 or more cases per group, t-tests and ANOVA tolerate non-normal data; with fewer, prefer the rank-based test if the data are clearly skewed (Lumley et al. (2002)).

Info Effect size. $|d| = 0.79$ is medium. The sign of d follows the order of the groups: $d = -0.79$ means Yes scored higher than No.

Rule: Cohen's d : about 0.2 small, 0.5 medium, 0.8 large (Cohen (1988)).

Group descriptives					
Group	n	Mean	SD	Median	Skewness
No	854	2637.02	1893.02	2330.50	1.12
Yes	646	4389.38	2591.22	3949.00	0.79

t-tests (two-sided)					
Test	t	df	p	Mean difference	95% CI
Welch's t (recommended)	-14.51	1131.47	< .001	-1752.36	[-1989.36, -1515.37]
Student's t (equal variances)	-15.13	1498	< .001	-1752.36	[-1979.50, -1525.22]

Figure 31. The independent-samples t-test of Annual_spend by Loyalty_member.

Table 16

Annual_spend by Loyalty_member

Group	n	Mean	SD	Median	Skewness
No	854	2637.02	1893.02	2330.50	1.12
Yes	646	4389.38	2591.22	3949.00	0.79

What it means for our example

Members spend 1752.36 more per year on average. Levene's test shows unequal variances ($F(1, 1498) = 70.54, p < .001$), so Welch's t-test is read: $t(1131.47) = -14.51, p < .001$, 95% CI of the difference $[-1989.36, -1515.37]$, $d = -0.79$ (a medium effect, bordering on large). SaNaML's effect-size line puts it this way: " $|d| = 0.79$ is medium. The sign of d follows the order of the groups: $d = -0.79$ means Yes scored higher than No." The negative sign therefore says nothing about the direction of the effect by itself; read it from the two means. The rank-based Mann-Whitney test agrees, $z = -14.09, p < .001$, rank-biserial $r = -.42$. On the logarithmic scale for buyers (Annual_spend_ln) the result is the same: members 8.24 against 7.76, $t(1371.00) = -14.35, p < .001, d = -0.76$, and the variances are equal (Levene $p = .104$).

Concept: when to read the rank-based test

SaNaML shows Welch's t-test and the Mann-Whitney U test side by side. With 30 or more customers per group, tests of means tolerate skewed data (Lumley et al., 2002), so here both agree. With small groups and strong skew, trust the rank-based test, and report medians.

How to report it

"Loyalty members spent more per year ($M = 4389.38$, $SD = 2591.22$) than non-members ($M = 2637.02$, $SD = 1893.02$), Welch's $t(1131.47) = -14.51$, $p < .001$, $d = -0.79$, 95% CI $[-0.90, -0.68]$."

5.5 Compare spend between three or more groups

Choose **Analyses** → **One-way ANOVA**, **Annual_spend** as the **Score to compare** and **Main_channel** as the **Groups** (Figure 32).

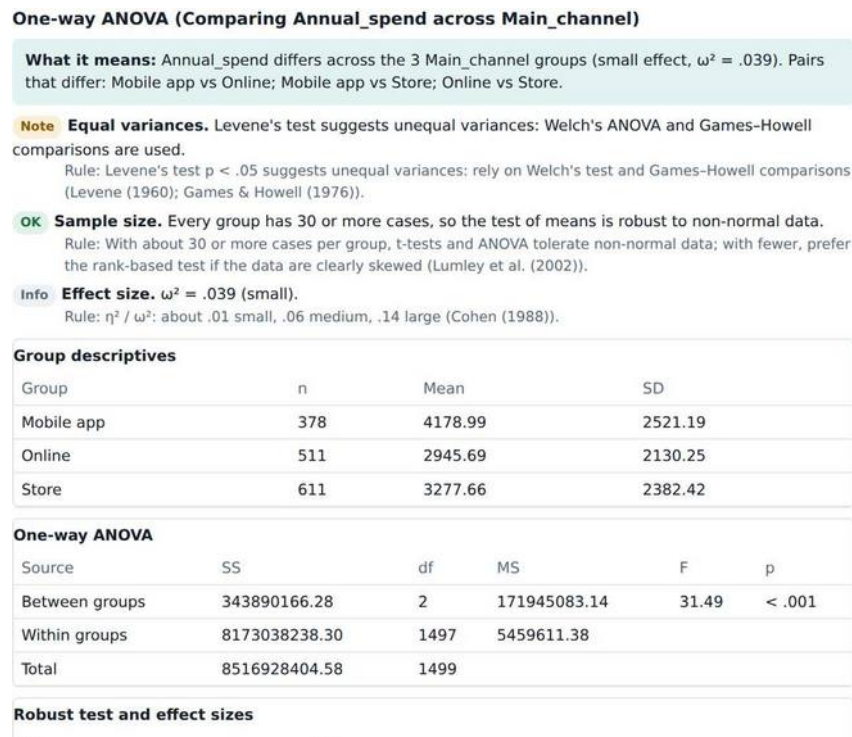


Figure 32. One-way ANOVA of Annual_spend across the three main channels.

Table 17

Annual_spend by Main_channel

Channel	n	Mean	SD
Mobile app	378	4178.99	2521.19
Online	511	2945.69	2130.25
Store	611	3277.66	2382.42

What it means for our example

Spend differs across channels (small effect, $\omega^2 = .039$). Levene's test shows unequal variances ($F(2, 1497) = 5.20$, $p = .006$), so Welch's ANOVA, $F(2, 899.44) = 29.87$, $p < .001$, and the Games-Howell comparisons are read. All three pairs differ: app customers spend 1233.30 more than online customers (95% CI $[856.83, 1609.77]$) and 901.33 more than store customers ($[521.91, 1280.75]$); online customers spend 331.97 less than store customers (adjusted $p = .037$). The Kruskal-Wallis test agrees, $H(2) = 65.95$, $p < .001$, $\epsilon^2 = .044$.

Now compare satisfaction across channels: no difference, $F(2, 1412) = 1.55$, $p = .213$, $\omega^2 = .001$ (Kruskal–Wallis $H(2) = 2.38$, $p = .304$). This matches the overlapping intervals in Figure 7.

Annual spend does not differ between regions ($F(4, 1495) = 0.22$, $p = .926$, $\omega^2 = -.002$; Kruskal–Wallis $H(4) = 0.50$, $p = .974$) or genders ($F(2, 1497) = 0.50$, $p = .606$), a useful reminder that not every customer attribute matters. Only 30 customers have Gender "other", so comparisons involving that group are imprecise.

Cut-off values: ANOVA

Levene's test $p < .05$: rely on Welch's test and the Games–Howell comparisons (Levene, 1960; Games & Howell, 1976). **ω^2 and η^2 :** about .01 small, .06 medium, .14 large (Cohen, 1988).

How to report it

"Annual spend differed by main channel, Welch's $F(2, 899.44) = 29.87$, $p < .001$, $\omega^2 = .039$; customers who mainly use the app spent most ($M = 4178.99$), online customers least ($M = 2945.69$)."

Exercises

Exercise: spend

1. Regress Annual_spend_ln on Income_k and Visits_per_month only. What happens to R^2 ? (Compare with section 5.2.)
1. Compare Visits_per_month between channels with a one-way ANOVA. Do app customers visit more? Think about what that says about the channel differences in spend.
1. Planted answer for the chapter: Annual_spend rises with income, satisfaction, loyalty membership and above all with visits (section 5.1), and loyalty members spend more (t-test, $d = -0.79$).

6. Predicting churn and response

Churn means that a customer leaves within the year (Churned = Yes, 250 of 1500 customers, 16.7%). A model that predicts churn lets a firm focus retention effort. SaNaML offers two routes, and this chapter uses both:

- **Logistic regression (Analyses)** explains which characteristics go with a higher chance of churn, with odds ratios and tests.
- **Machine learning** compares several prediction models on customers they have not seen, explains the best one and checks whether its probabilities can be trusted.

Concept: explanation and prediction

A regression is judged by how well it explains the data it was estimated on. Machine learning asks a different question: how well does a model predict customers it has never seen? If flexible models (random forest, boosting) do not beat a plain logistic regression, the relationships are probably simple, and the regression is the model to report.

6.1 Which variables may be used as predictors?

A prediction is useful only if the predictors are known *before* the outcome. Two kinds of variable must stay out of a churn model:

- **Variables measured over the same year or later than the outcome.** Annual_spend covers the year in which customers left, so a customer who left in March has low annual spend because they left: the predictor partly *contains* the outcome (*leakage*). The two monthly spend variables were measured around the campaign.
- **Variables that are results of a campaign you want to evaluate separately**, here Campaign_group and Responded.

The predictors we use for churn are the profile (Age, Gender, Region, Income_k), the relationship (Tenure_months, Loyalty_member, Main_channel), behaviour (Visits_per_month, Avg_basket, Discount_share, Complaints_12m) and the two ratings (Satisfaction_1to10, Recommend_0to10). CustomerID is not a predictor: the Machine learning screen leaves a text identifier out on its own (section 6.3).

Note: leakage

A model with leakage looks excellent and fails in use. SaNaML guards against some forms (a predictor almost identical to the outcome, a respondent number or text identifier, a construct that contains the outcome), but it cannot know when a variable was measured. That is your responsibility.

6.2 Logistic regression for churn

Choose **Analyses** → **Correlation and regression analysis** → **Binary logistic regression (Which variables predict a yes/no outcome?)**. Choose **Churned** as the outcome (the event, coded 1, is

"Yes") and tick as predictors Satisfaction_1to10, Complaints_12m, Tenure_months and Loyalty_member (Figure 33).

Binary logistic regression (Predicting Churned = "Yes")

What it means: The model predicts "Yes" better than chance (Nagelkerke $R^2 = .19$, AUC = .75, acceptable). Loyalty_member: Yes (vs No): each unit divides the odds of "Yes" by 1.45; Complaints_12m: each unit multiplies the odds of "Yes" by 1.39; Satisfaction_1to10: each unit divides the odds of "Yes" by 1.64.

OK Events per variable. EPV = 58.5.

Rule: At least 10 respondents in the smaller outcome group per coefficient (EPV ≥ 10) (Peduzzi et al. (1996)).

OK Calibration. Hosmer-Lemeshow $\chi^2(8) = 10.68$, $p = .221$.

Rule: Hosmer-Lemeshow $p \geq .05$: no evidence of poor fit (Hosmer et al. (2013)).

OK Discrimination. AUC = .754 (acceptable).

Rule: AUC .7-.8 acceptable, .8-.9 excellent, $\geq .9$ outstanding (Hosmer et al. (2013)).

Model fit

χ^2	df	p	Nagelkerke R^2	McFadden R^2	Tjur R^2	AUC	95% CI of AUC
169.84	4	< .001	.191	.134	.125	.754	[.72, .79]

Coefficients

Predictor	B	SE	Wald	p	Odds ratio	95% CI of OR
(Intercept)	1.493	0.362	17.03	< .001	4.450	[2.190, 9.043]
Tenure_months	-0.008	0.004	3.72	.054	0.992	[0.983, 1.000]
Loyalty_member: Yes (vs No)	-0.374	0.172	4.73	.030	0.688	[0.491, 0.964]
Complaints_12m	0.329	0.073	20.52	< .001	1.390	[1.205, 1.603]
Satisfaction_1to10	-0.498	0.057	75.33	< .001	0.608	[0.543, 0.680]

"Yes" = 1, "No" = 0. Categorical predictors are dummy-coded against their first group.

Classification at a 0.5 cut-off

Figure 33. The logistic regression of Churned: what it means, the checks and the model fit.

Concept: odds ratio

An odds ratio (OR) is the factor by which the odds of the event are multiplied when the predictor increases by one unit (or when a customer belongs to the group rather than the reference group). An OR of 1 means no effect; above 1 the event becomes more likely; below 1 less likely. The odds of an event are its probability divided by one minus its probability.

Table 18

Logistic regression of Churned (N = 1415 customers with a satisfaction rating)

Predictor	Odds ratio	p	95% CI of OR
Tenure_months (per month)	0.992	.054	[0.983, 1.000]
Loyalty_member: Yes (vs No)	0.688	.030	[0.491, 0.964]
Complaints_12m (per complaint)	1.390	< .001	[1.205, 1.603]
Satisfaction_1to10 (per point)	0.608	< .001	[0.543, 0.680]

What it means for our example

Each additional complaint multiplies the odds of leaving by 1.39, each additional satisfaction point multiplies them by 0.61 (the odds fall by about 39%), and loyalty members have 0.69 times the odds of non-members. Longer tenure points the same way but its interval just reaches 1 ($p = .054$). The model beats chance ($\chi^2(4) = 169.84$, $p < .001$; $AUC = .754$, 95% CI [.72, .79], "acceptable"; Nagelkerke $R^2 = .19$). The checks are fine: 58.5 events per variable (EPV, at least 10 needed; Peduzzi et al., 1996) and the Hosmer–Lemeshow test shows no poor fit, $\chi^2(8) = 10.68$, $p = .221$.

Cut-off values: logistic regression

EPV ≥ 10 (Peduzzi et al., 1996). **AUC** .70–.80 acceptable, .80–.90 excellent, $\geq .90$ outstanding (Hosmer et al., 2013). **Hosmer–Lemeshow** $p \geq .05$: no evidence of poor fit.

Adding Age, Region, Main_channel, Visits_per_month and Discount_share (nine predictors, 13 coefficients) changes little: Nagelkerke $R^2 = .21$, $AUC = .766$ (95% CI [.73, .80]). Complaints (OR = 1.40) and satisfaction (OR = 0.61) keep their effects; age, channel, visits and discount share do not matter; Region does: relative to Central, the odds of churn are lower in the East (OR = 0.59), North (0.55), South (0.55) and West (0.48), all $p < .05$.

The accuracy trap

The classification table at the bottom of the result shows what the model does at a 0.5 cut-off (a customer is predicted to churn when the predicted probability is above 0.5):

Table 19

Classification of churn at a 0.5 cut-off (model with four predictors)

Observed	Predicted No	Predicted Yes	% correct
No	1159	22	98.1
Yes	216	18	7.7

Overall, 83.2% are classified correctly, and SaNaML adds: "Guessing the larger group for everyone would be 83.5% correct." **The model is less accurate than a model that predicts "no churn" for everyone.** It finds only 18 of the 234 customers who left.

Concept: precision, recall and why accuracy misleads

For a rare outcome, the share of correct predictions is dominated by the common group. Use instead:

- **Recall (sensitivity):** of the customers who really left, the share the model found.
- **Precision:** of the customers the model flagged, the share who really left.
- **Specificity:** of the customers who stayed, the share the model left alone.

A single cut-off trades recall against precision: lowering the cut-off finds more leavers but also flags more customers who would have stayed. The area under the curve (AUC) summarises ranking quality over all cut-offs and does not depend on the cut-off. Section 6.5 shows the three measures for the machine learning model at every cut-off.

What it means for our example

At the 0.5 cut-off the logistic regression finds only 18 of the 234 customers who left; the other 216 are missed. For a retention campaign that is the wrong cut-off: the goal is to find many customers at risk, and a flagged customer who would have stayed costs only the price of a contact. Section 6.5 shows what lower cut-offs do.

6.3 Machine learning: set up the prediction

Click **Machine learning** in the menu. The screen has three set-up cards and, after the run, the results.

1. In 1 · *Prediction task*, choose **Churned (yes/no: No / Yes)** as the **Outcome to predict**. SaNaML sets the task (binary classification) and ticks every other variable as a predictor. **Which answer is the event (coded 1)?** is **Yes**.
2. **Untick** Annual_spend, Monthly_spend_before, Monthly_spend_after, Campaign_group and Responded (section 6.1) (Figure 34). The box of CustomerID is already unticked: SaNaML recognises it as an identifier (see the note below).
3. Read the checks under the predictors: SaNaML notes "The answers are unbalanced (17% vs 83%). Read AUC and balanced accuracy rather than plain accuracy." and states that missing values, standardising and tuning are learned from the training customers of each fold only (nested cross-validation).
4. In 2 · *Validation plan* choose **5-fold cross-validation, no repeats (Quick look)** for exploring; use the recommended repeated cross-validation for results you report. The **Main measure** is AUC for yes/no outcomes, and the **Random seed** (2026) makes every split reproducible.
5. In 3 · *Models* keep the default models (the stacked ensemble is off) and click **Run the comparison**. A progress bar shows the steps.

1 · Prediction task (What to predict, from what)

Outcome to predict

Churned (yes/no: No / Yes)

A score (regression), a variable with two answers (binary classification) or one with 3–10 categories (multiclass classification), for example a construct score such as behavioural intention.

Which answer is the event (coded 1)?

☒ **Yes**
Predict the probability of "Yes".

☐ **No**
Predict the probability of "No".

Predictors

Select all Select none

☐ CustomerID (groups) ☒ Age ☒ Gender (groups) ☒ Region (groups) ☒ Income_k

☒ Tenure_months ☒ Loyalty_member (groups) ☒ Main_channel (groups) ☒ Visits_per_month ☒ Avg_basket

☒ Discount_share ☒ Complaints_12m ☒ Satisfaction_1to10 ☒ Recommend_0to10 ☐ Annual_spend

☐ Campaign_group (groups) ☐ Responded (groups) ☐ Monthly_spend_before ☐ Monthly_spend_after

Info: **Unbalanced answers.** The answers are unbalanced (17% vs 83%). Read AUC and balanced accuracy rather than plain accuracy.

OK: **Sample size.** 1500 respondents.

Info: **Leakage protection.** Filling in missing values, standardising and tuning are all learned from the training rows of each fold only, so the test rows never influence the model (nested cross-validation).

Figure 34. 1 · Prediction task: the outcome, the predictors and the checks.

Note: the program keeps the customer ID out for you

SaNaML recognises a text identifier such as C0001 (every customer has a different value) and leaves its box unticked in the list of predictors, so there is nothing to untick. If you tick it, SaNaML answers: "CustomerID looks like a respondent identifier (every value different). A model could learn the order in which data were collected, or memorise individual respondents, rather than anything general about them. It is always left out of the model, whether or not its box is ticked." The quick look on 1500 customers took 345 seconds (about six minutes) on the computer used for this manual; speed depends on your computer.

Concept: cross-validation

The customers are split into k parts (*folds*). Each model is trained on $k - 1$ parts and tested on the remaining one, k times over, so every customer is predicted once by a model that did not see them. SaNaML keeps the share of churners the same in every fold (stratified folds) and tunes each model by an inner cross-validation inside every training set (nested cross-validation), so that the test customers never influence the model (Kuhn & Johnson, 2013).

6.4 Machine learning: compare the models

Out-of-sample performance (nested cross-validation)									
Model	AUC M (SD)	95% CI	Accuracy	Balanced accuracy	F1	MCC	Brier score	Δ vs best (p)	
Logistic regression	.755 (.024)	[.711, .799]	.833	.553	.207	.188	0.122	best	
Elastic net	.754 (.019)	[.719, .789]	.833	.517	.076	.081	0.124	.000 (.959)	
XGBoost	.745 (.015)	[.718, .772]	.835	.523	.100	.143	0.123	-.010 (.477)	
LightGBM	.743 (.023)	[.701, .786]	.831	.528	.122	.127	0.124	-.011 (.442)	
Random forest	.722 (.034)	[.659, .785]	.837	.538	.149	.173	0.125	-.033 (.061)	
Decision tree (CART)	.691 (.023)	[.648, .733]	.821	.552	.215	.163	0.134	-.064 (.018)	
Null model	.500 (.000)	[.500, .500]	.833	.500	.000	.000	0.139	-.255 (< .001)	

Figure 35. Out-of-sample performance of the seven models (5-fold cross-validation).

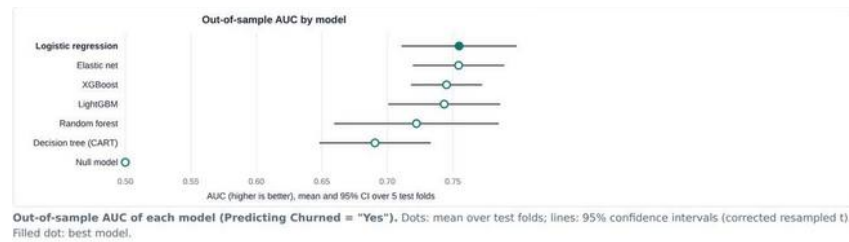


Figure 36. Out-of-sample AUC of each model (dots: mean over the test folds; lines: 95% confidence intervals).

Table 20

Out-of-sample prediction of Churned (5-fold cross-validation)

Model	AUC M (SD)	95% CI	Accuracy	Balanced accuracy	F1	MCC	Brier	Δ vs best (p)
Logistic regression	.755 (.024)	[.711, .799]	.833	.553	.207	.188	0.122	best
Elastic net	.754 (.019)	[.719, .789]	.833	.517	.076	.081	0.124	.000 (.959)
XGBoost	.745 (.015)	[.718, .772]	.835	.523	.100	.143	0.123	-.010 (.477)
LightGBM	.743 (.023)	[.701, .786]	.831	.528	.122	.127	0.124	-.011 (.442)
Random forest	.722 (.034)	[.659, .785]	.837	.538	.149	.173	0.125	-.033 (.061)
Decision tree (CART)	.691 (.023)	[.648, .733]	.821	.552	.215	.163	0.134	-.064 (.018)
Null model	.500 (.000)	[.500, .500]	.833	.500	.000	.000	0.139	-.255 (< .001)

What it means for our example

Logistic regression predicts churn best on customers it has not seen (AUC = .755, 95% CI [.711, .799]), and no flexible model beats it: elastic net, XGBoost and LightGBM are statistically indistinguishable ($p = .959$, .477 and .442), random forest is not significantly worse ($p = .061$) and the single decision tree is significantly worse ($p = .018$). Churn in this file is therefore explained by simple, additive relationships, and the logistic regression of section 6.2 is the model to report. Only the null model, which predicts the average for everyone, is clearly worse.

Now look at the accuracy column: every model, including the null model, is about 83% accurate (.821 to .837), because 83% of customers stay. Accuracy does not separate the models at all. The AUC does.

Table 21

Measures reported for a yes/no outcome

Measure	What it tells you
AUC	The chance that a randomly chosen customer who left gets a higher predicted probability than a randomly chosen customer who stayed. 0.5 is chance, 1 is perfect. Independent of the cut-off; the main measure.
Accuracy	The share of customers classified correctly at the 0.5 cut-off. Misleading for rare outcomes.
Balanced accuracy	The average of recall and specificity: .50 for a model that predicts one group for everyone.

Measure	What it tells you
F1	The harmonic mean of precision and recall for the churners.
MCC	Matthews correlation coefficient: a correlation between prediction and truth that uses all four cells of the classification table (0 = chance, 1 = perfect).
Brier score	The mean squared error of the predicted probabilities (lower is better); the null model sets the benchmark (0.139).

Note: small samples and flexible models

Flexible models need many more customers than linear models before they pay off (van der Ploeg et al., 2014). With a few hundred or a few thousand customers, "no better than logistic regression" is a common and useful result. The Quick look is noisy; use the repeated cross-validation for the results you report.

What the program does about imbalanced outcomes

- It **warns** you under the predictors ("The answers are unbalanced (17% vs 83%)") and warns about a rare answer when fewer than 50 respondents (borderline) or 20 respondents (check) have it.
- It makes **AUC** the main measure by default, and reports balanced accuracy, F1, MCC and the Brier score next to accuracy.
- It always includes the **null model**, which shows what plain accuracy gives without any information.
- It keeps the **share of churners equal in every fold** (stratified folds).
- It lets you **choose the cut-off** and shows precision, recall, specificity and F1 at every cut-off for the model you explain, with the average precision (section 6.5).
- It does **not** reweight or resample the classes, and it does not report precision and recall for every model at once.

6.5 Choosing a cut-off: precision and recall

The prediction models give each customer a *probability* of churn. A decision needs a *cut-off*: contact every customer whose probability is above it. The best cut-off depends on costs, not on statistics (Fawcett, 2006).

1. Estimate the **cost of contacting** a customer who would have stayed (an offer, a call, a discount given away).
2. Estimate the **value of keeping** a customer who would have left, multiplied by the chance that the retention action works.
3. Contact a customer when the probability of churn times the value kept exceeds the cost of contacting. In symbols: contact when $p \times V > C$, that is, when the churn probability p is above $C \div V$. A cut-off of 0.20, for example, is the right one when the value kept is five times the cost of a contact.

SaNaML helps you to see what a cut-off does. For a yes/no outcome the Machine learning results end with the card *Results · Cut-offs, precision and recall (What happens when probabilities become yes or no?)*. It evaluates the model chosen in [Explain this model](#) (here the logistic regression) on predictions for customers the model did not see. Type a number between 0 and 1 in **Cut-off (probability, between 0 and 1)**; the card starts at 0.5 (Figure 37).

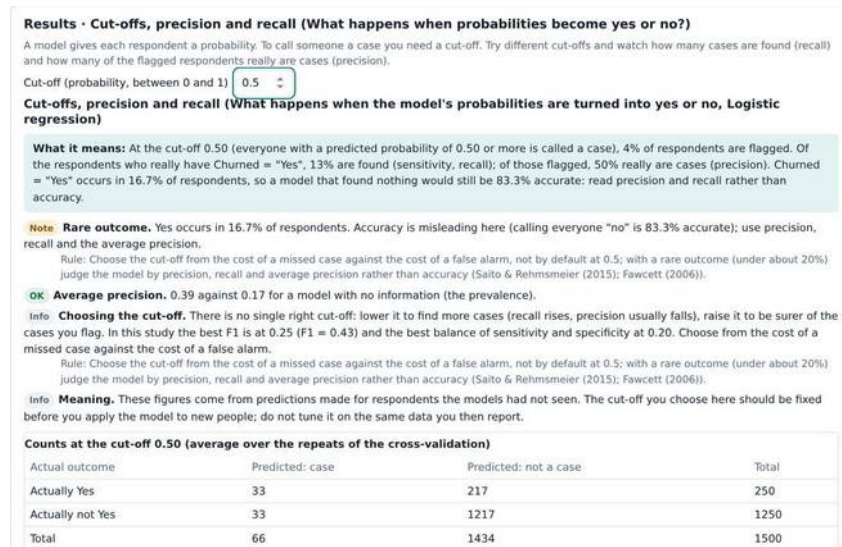


Figure 37. Cut-offs, precision and recall at the cut-off 0.5.

What it means for our example: the cut-off 0.5

SaNaML writes: "At the cut-off 0.50 (everyone with a predicted probability of 0.50 or more is called a case), 4% of respondents are flagged. Of the respondents who really have Churned = "Yes", 13% are found (sensitivity, recall); of those flagged, 50% really are cases (precision)." The count table shows the same in customers: of the 250 customers who left, 33 are found and 217 are missed; of the 66 customers flagged, 33 really leave and 33 do not. The card adds that the outcome is rare (Yes occurs in 16.7% of customers, so calling everyone "no" is 83.3% accurate) and that the average precision is 0.39 against 0.17 for a model with no information, which it marks OK.

Now type **0.2** (Figure 38). The model flags more customers and finds more of the leavers.

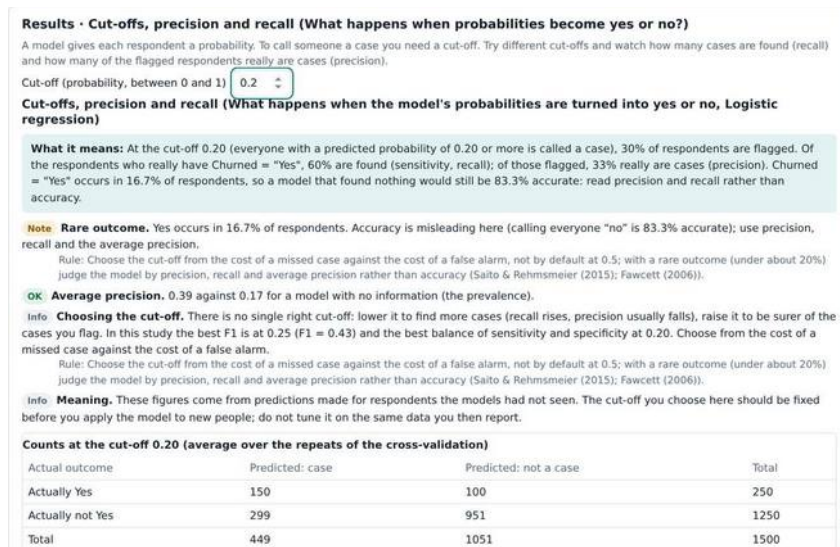


Figure 38. The same card at the cut-off 0.2.

Table 22

What the cut-off does (logistic regression, Churned, 5-fold cross-validation)

Cut-off	Flagged	Recall	Precision	Specificity	F1	Accuracy
0.10	58%	0.86	0.25	0.48	0.38	0.54
0.20	30%	0.60	0.33	0.76	0.43	0.73
0.30	15%	0.38	0.41	0.89	0.40	0.81
0.50	4%	0.13	0.50	0.97	0.21	0.83

What it means for our example: the cut-off 0.2

At 0.20, 30% of the customers are flagged (449 of 1500). Of the 250 customers who left, 150 are found (recall 0.60) and 100 are missed; of the 449 flagged customers 150 really leave (precision 0.33) and 299 do not. Compared with the cut-off 0.5, the program finds 150 instead of 33 of the leavers, and the price is 299 instead of 33 false alarms. The accuracy falls from .83 to .73, which shows once more why accuracy is the wrong measure for a rare outcome. SaNaML's own hint is that, among the cut-offs it tries, the best F1 is at 0.25 (F1 = 0.43) and the best balance of sensitivity and specificity at 0.20.

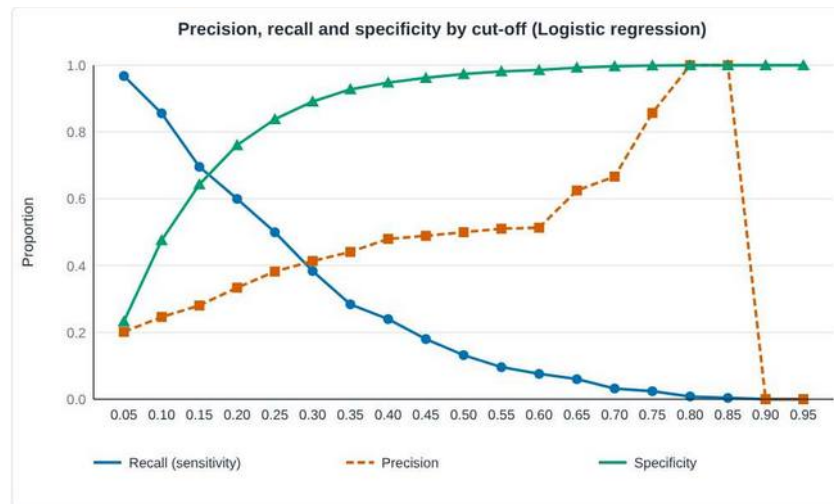


Figure 39. Precision, recall and specificity of the logistic regression at every cut-off.

Concept: precision and recall move in opposite directions

A high cut-off flags few customers, mostly true leavers (high precision, low recall); a low cut-off flags many customers and finds most leavers, but most flagged customers would have stayed (low precision, high recall). In Figure 39 the recall curve falls and the specificity curve rises as the cut-off grows, and the precision curve rises with it. Read the far right of the chart with care: at cut-offs of 0.80 and above fewer than half a percent of the customers are flagged, so a precision of 1.00 there rests on a handful of customers, and SaNaML draws an undefined precision (nobody flagged) as 0. A model with no information has a precision equal to the prevalence (0.17); the average precision summarises the whole precision–recall curve (Saito & Rehmsmeier, 2015; He & Garcia, 2009).

Note: choose the cut-off before you look at the result you report

The cut-off that SaNaML suggests comes from the same predictions that it evaluates. Fix the cut-off from your costs (the formula above) before applying the model to new customers, and do not tune it on the data you then report. The recall and precision of a rare outcome are also noisy: the Quick look uses one 5-fold cross-validation, and the recommended repeated cross-validation gives more stable figures.

Because the cut-off is a business decision, the useful output of a prediction model is a *good ranking and trustworthy probabilities*. Calibration (section 6.7) tells you whether the probabilities can be trusted.

6.6 Machine learning: explain the best model

Explain this model chooses the model to explain; the options mark the best model. For churn, the explanations of the logistic regression are shown.

Importance of each predictor				
Predictor	Mean SHAP	SHAP rank	Direction	Permutation importance M (SD)
Satisfaction_1to10	0.492	1	higher Satisfaction_1to10 → lower the chance of Churned = "Yes"	0.096 (0.009)
Complaints_12m	0.298	2	higher Complaints_12m → higher the chance of Churned = "Yes"	0.040 (0.018)
Loyalty_member	0.218	3	depends on the category	0.009 (0.007)
Region	0.195	4	depends on the category	0.010 (0.009)
Tenure_months	0.166	5	higher Tenure_months → lower the chance of Churned = "Yes"	0.008 (0.008)
Recommend_0to10	0.117	6	higher Recommend_0to10 → lower the chance of Churned = "Yes"	0.008 (0.007)
Gender	0.117	7	depends on the category	0.008 (0.008)
Discount_share	0.063	8	higher Discount_share → higher the chance of Churned = "Yes"	0.002 (0.001)
Main_channel	0.061	9	depends on the category	-0.001 (0.004)
Income_k	0.044	10	higher Income_k → lower the chance of Churned = "Yes"	0.001 (0.002)
Visits_per_month	0.023	11	higher Visits_per_month → lower the chance of Churned = "Yes"	0.001 (0.002)
Age	0.010	12	higher Age → higher the chance of Churned = "Yes"	-0.001 (0.003)
Avg_basket	0.004	13	higher Avg_basket → lower the chance of Churned = "Yes"	-0.004 (0.005)

Figure 40. Importance of each predictor: SHAP and permutation importance.

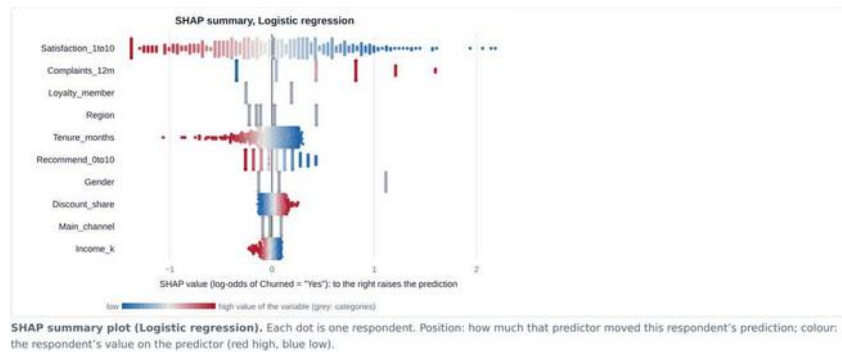


Figure 41. SHAP summary plot: one dot per customer and predictor.

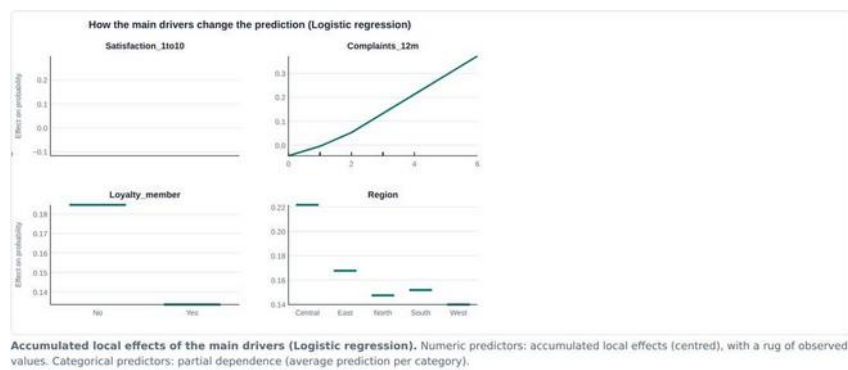


Figure 42. Accumulated local effects of the main drivers.

Table 23

The strongest drivers of churn in the logistic regression (importance on held-out customers)

Predictor	Mean abs. SHAP	Direction	Permutation importance M (SD)	Rank (SHAP / permutation)
Satisfaction_1to10	0.492	higher satisfaction, lower chance of churn	0.096 (0.009)	1 / 1
Complaints_12m	0.298	more complaints, higher chance	0.040 (0.018)	2 / 2
Loyalty_member	0.218	depends on the category	0.009 (0.007)	3 / 4
Region	0.195	depends on the category	0.010 (0.009)	4 / 3
Tenure_months	0.166	longer tenure, lower chance	0.008 (0.008)	5 / 5

Concept: SHAP, permutation importance and ALE

SHAP values split each customer's prediction into the contributions of the predictors; the mean absolute value shows how much a predictor moves predictions on average (here in log-odds of churn). *Permutation importance* shuffles one predictor in the held-out customers and measures the loss in AUC. *Accumulated local effects* (ALE) show how the predicted probability changes across the range of one predictor while the other predictors stay realistic. These tools describe what the model learned, **not causal effects** (Molnar, 2022). Correlated predictors share credit, and ALE is preferred to partial dependence when predictors are correlated (Apley & Zhu, 2020).

What it means for our example

Satisfaction is the main driver (mean $|\text{SHAP}| = 0.492$), followed by complaints (0.298), loyalty membership (0.218), region (0.195) and tenure (0.166). The two importance methods agree on the top two; satisfaction loses 0.096 AUC when shuffled and complaints 0.040. The ALE table shows the shape: the predicted churn probability is 0.299 above average at a satisfaction of 2.30 and 0.117 below average at 10.00, a range of about 0.42, and with complaints it rises from 0.045 below average at 0 complaints to 0.372 above average at 6. Loyalty members have a lower average predicted churn probability (0.13 against 0.18), and the East, North, South and West regions lower than Central (0.14 to 0.17 against 0.22). Main channel, age and average basket add nothing: shuffling them did not hurt the predictions. Satisfaction ranks first in all six models and complaints second or third, which makes these two drivers credible; the rank of loyalty membership and region varies between models.

How to report it

"Seven prediction models were compared with 5-fold cross-validation (5 train/test splits), with every model tuned by an inner 5-fold cross-validation inside each training set (nested cross-validation; seed 2026). Logistic regression performed best (AUC = .755, 95% CI [.711, .799]; balanced accuracy = .553)."

6.7 Calibration: can the probabilities be trusted?

Under *Results · Precision and calibration*, SaNaML checks whether the predicted probabilities mean what they say (Figure 43).

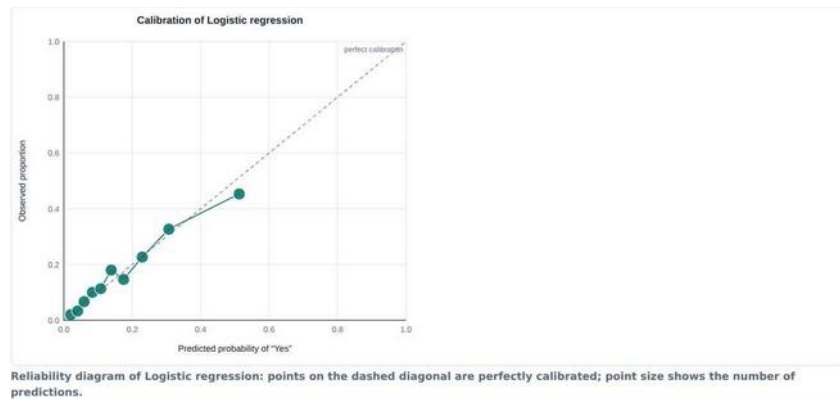


Figure 43. Reliability diagram: predicted probability against the observed proportion of churners.

What it means for our example

The calibration slope is 0.89 (95% CI [0.75, 1.04]) and calibration-in-the-large is -0.01 (95% CI [-0.15, 0.14]); the predictions are about as spread out as they should be and there is no systematic over- or under-prediction. In the reliability table, customers in the tenth with the lowest predicted risk (mean .020) churned at .020, and customers in the tenth with the highest predicted risk (mean .513) churned at .453. The highest-risk tenth of customers (150 predictions per bin) therefore churns at .453, against the overall rate of 16.7% (.167). A predicted 30% can be read as roughly a 30% chance, which is what a cost-based cut-off needs.

Cut-off values: calibration

Calibration slope close to 1 (below 1: predictions too extreme, a sign of overfitting; above 1: too timid) and calibration-in-the-large close to 0 (Van Calster et al., 2019).

Note: what this version cannot show

SaNaML 0.10.2 shows the cut-off table but does not list the predicted probability of each customer, so it cannot score a customer list. Use the comparison to decide *which method* to use and what drives churn; apply the chosen model in the program or tool that holds your customer database. Do not base decisions about individual customers on a model without human review, and think about whether using variables such as gender or region for targeting is fair and permitted.

6.8 Predicting response to the campaign

The same steps predict whether a customer responded to the campaign. On **Machine learning**, choose **Responded (yes/no: No / Yes)** as the outcome and untick Annual_spend, Monthly_spend_before, Monthly_spend_after and Churned (CustomerID is unticked by default, section 6.3). Keep **Campaign_group**: customers were assigned to it at random, and it was known before they responded. Use the Quick look and run the comparison.

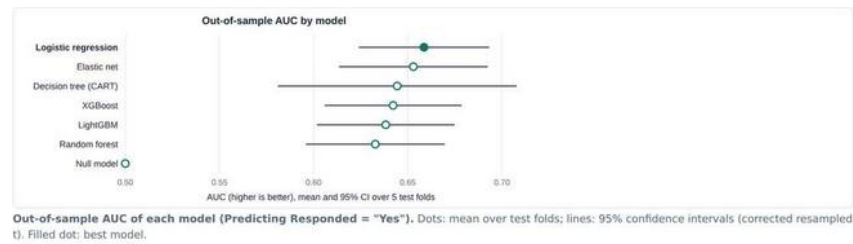


Figure 44. Out-of-sample AUC of each model when predicting Responded.

Table 24

Out-of-sample prediction of Responded (5-fold cross-validation)

Model	AUC M (SD)	95% CI	Accuracy	Balanced accuracy	F1	MCC	Brier	Δ vs best (p)
Logistic regression	.659 (.019)	[.624, .693]	.671	.586	.388	.209	0.212	best
Elastic net	.653 (.021)	[.613, .693]	.661	.559	.291	.162	0.214	-.006 (.565)
Decision tree (CART)	.644 (.034)	[.581, .708]	.673	.598	.426	.227	0.213	-.014 (.433)
XGBoost	.642 (.020)	[.606, .679]	.661	.560	.312	.168	0.216	-.016 (.242)
LightGBM	.638 (.020)	[.602, .675]	.659	.561	.319	.165	0.216	-.020 (.121)
Random forest	.633 (.020)	[.596, .670]	.641	.550	.321	.123	0.219	-.026 (.027)
Null model	.500 (.000)	[.500, .500]	.649	.500	.000	.000	0.228	-.159 (< .001)

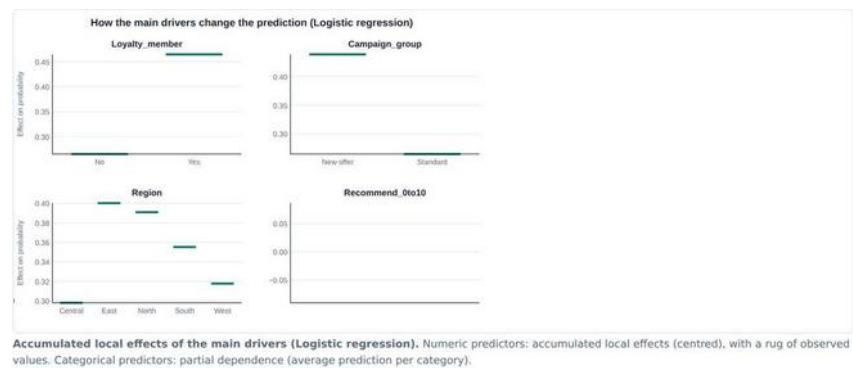


Figure 45. Accumulated local effects of the main drivers of response.

What it means for our example

Logistic regression is again the best model (AUC = .659, 95% CI [.624, .693]), no other model is significantly better, and the random forest is significantly worse ($p = .027$). The AUC is below .70, and SaNaML marks the discrimination as *Note*: knowing the offer, loyalty status, region and the other variables separates responders from non-responders only modestly. The accuracy of the null model, which predicts "No" for everyone, is .649, close to that of the best model (.671), so accuracy tells us little here too (35% of the customers responded).

The strongest drivers are loyalty membership (mean $|SHAP| = 0.460$) and the campaign group (0.415); permutation importance puts them first and second as well (loss in AUC 0.087 and 0.070), and all six models rank them in the same two places. The average predicted probability of responding is 0.46 for loyalty members against 0.26 for non-members, and 0.44 after the New offer against 0.26 after the Standard offer. The regions differ somewhat (Central 0.30, East 0.40, North 0.39, South 0.36, West 0.32), and higher scores on Recommend_0to10 go with a higher predicted probability (about 0.18 across its range).

What it means for our example: calibration

The calibration slope is 0.79 (95% CI [0.63, 0.95]) and SaNaML marks it *Check*: the predictions are too extreme (high predictions too high, low predictions too low), the usual sign of overfitting. In the reliability table, the tenth of customers with the lowest predicted response (mean .134) responded at .200, and the tenth with the highest predicted response (mean .648) responded at .580. Read the probabilities as a ranking, not as exact chances. Compare this with the churn model (slope 0.89, section 6.7): a model can discriminate and calibrate better for one outcome than for another.

Table 25

What the cut-off does when predicting Responded (logistic regression, 5-fold cross-validation)

Cut-off	Flagged	Recall	Precision	Specificity	F1	Accuracy
0.05	100%	1.00	0.35	0.00	0.52	0.35
0.30	58%	0.73	0.44	0.50	0.55	0.58
0.50	19%	0.30	0.56	0.87	0.39	0.67

What it means for our example: cut-offs for the response model

The *Cut-offs, precision and recall* card of this model (Table 25) gives a different picture from the churn model, because 35% of the customers respond: SaNaML rates the classes as reasonably balanced, and its average precision is 0.50 against 0.35 for a model with no information. At the cut-off 0.50, 19% of the customers are flagged; 158 of the 526 responders are found (recall 0.30) and 56% of the flagged customers respond (precision 0.56, 158 of 284). At 0.30, 58% are flagged, 385 of the 526 responders are found (recall 0.73) and the precision falls to 0.44 (385 of 876). The first row shows the baseline: flagging everybody finds all responders and has a precision of 0.35, the response rate. A cut-off is useful only to the extent that precision rises above that baseline. SaNaML's hint is that the best F1 is at 0.25 ($F1 = 0.55$) and the best balance of sensitivity and specificity at 0.30. The choice still depends on what a contact costs and what a response is worth (section 6.5).

Note: the offer effect is shown by the A/B test, not by the prediction

Which customers respond is a prediction question; whether the *offer* raised the response is a causal question that the randomised A/B test of section 4.1 answers. The model agrees (the New offer raises the predicted response from 0.26 to 0.44, in line with the observed 26% and 44%), but SaNaML does not offer uplift modelling, which asks which customers are moved by the offer.

How to report it

"Seven prediction models were compared with 5-fold cross-validation. Logistic regression performed best (AUC = .659, 95% CI [.624, .693]; balanced accuracy = .586); discrimination was modest and the predicted probabilities were somewhat too extreme (calibration slope 0.79, 95% CI [0.63, 0.95])."

Exercises

Exercise: churn and response

1. Predict Churned again after adding Annual_spend as a predictor. What happens to the AUC, and why should you not believe it? Think about leakage (section 6.1).
1. In the machine learning comparison of Churned, set the cut-off of the card *Cut-offs, precision and recall* to 0.30 and read the recall and the precision. Answer: recall 0.38 (96 of the 250 leavers) and precision 0.41 (96 of the 232 flagged customers); see Table 22 for the other cut-offs.
1. Compare the calibration of the churn model and the response model. Which one are you more willing to use as a probability? Answer: the churn model (slope 0.89, "OK") rather than the response model (slope 0.79, "Check").
1. Planted answer for the chapter: churn is driven by satisfaction, complaints, tenure and loyalty (section 6.2); response by the offer, loyalty and channel (section 6.8, logistic regression of Responded: loyalty members OR = 2.51, standard offer OR = 0.43, relative to the New offer).

7. Customer segmentation

Marketing often asks whether the customers fall into a few groups with different needs, so that each group can be treated differently. *Cluster analysis* looks for groups of customers who are similar to each other and different from the others. SaNaML's **Clustering** screen offers k-means and hierarchical clustering.

Concept: what cluster analysis does, and does not do

Cluster analysis always returns clusters, even when the data contain none. A solution is only useful if it is (a) reasonably separated (the *silhouette*), (b) stable when the sample changes (*stability*) and (c) interpretable and useful. The clusters describe this sample; they are not proof that customer "types" exist. Naming a cluster is a judgement by the analyst, not a result.

SaNaML standardises every variable before clustering (mean 0, standard deviation 1), so a variable measured in large units does not dominate. Customers with a missing value on any chosen variable are left out.

7.1 Segments with clear structure: segments_demo.csv

The file *segments_demo.csv* has 360 simulated shoppers rated on five attitude scales (Price_sensitivity, Brand_loyalty, Quality_focus, Online_affinity, Social_influence, each 1 to 7), with Age, Income_k, Gender and Channel (Mixed, Online, Store). Three segments were planted.

1. On **Data**, open *segments_demo.csv* (360 rows, 10 variables). Click **Clustering** in the menu.
2. In 1 • *Settings (What to cluster)*, tick the five attitude scales. **Untick Age and Income_k**: we want to *describe* the clusters with them, not build the clusters from them.
3. Open **Compare the clusters on other variables (optional)** and tick Age, Income_k, Gender and Channel.
4. Keep **Method** k-means, solutions from 2 to 8 clusters, **Show** the best silhouette, and tick **Also compute the gap statistic (slower)** (Figure 46).
5. Click **Find the clusters**.

1 • Settings (What to cluster)

Variables that describe the respondents (numeric)

Select all Select none

☒ Price_sensitivity ☒ Brand_loyalty ☒ Quality_focus ☒ Online_affinity ☒ Social_influence ☐ Age ☐ Income_k

Variables are standardised before clustering, so a variable on a 1-7 scale does not outweigh one on a 1-5 scale. Respondents with a missing value on any chosen variable are left out. Use construct or latent variable scores, or items measured the same way.

Method: k-means clustering (Groups around centres)

Compare solutions from 2 to 8 clusters Show the best silhouette (Recommended)

☒ Also compute the gap statistic (slower) Random seed 1

▼ Compare the clusters on other variables (optional)

Variables to describe the clusters

☒ Age ☒ Income_k ☒ Gender ☒ Channel

Find the clusters

Figure 46. The clustering settings for *segments_demo.csv*.

The result starts with a plain-language summary (Figure 47), then the checks, the table of solutions, the cluster means, the comparison with the other variables and the figures.

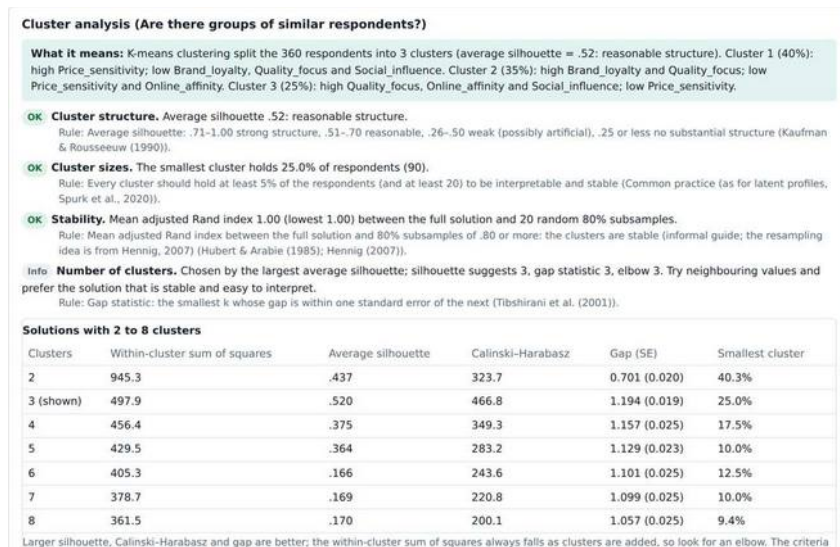
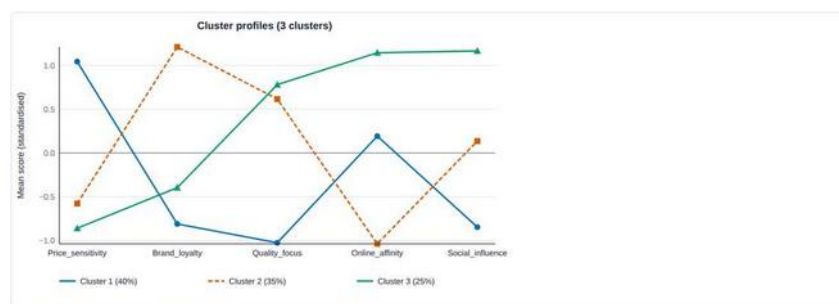
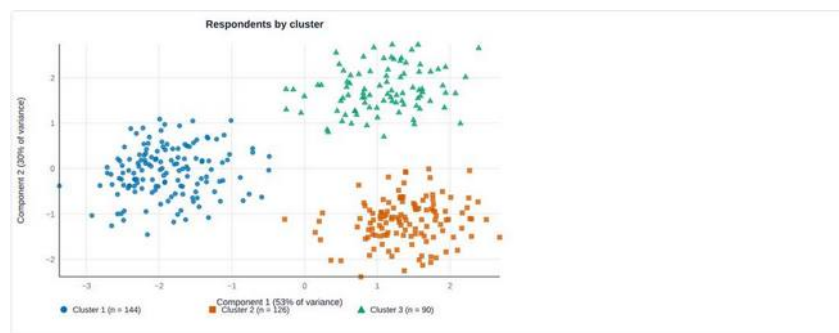


Figure 47. The start of the cluster report: the summary, the checks and the table of solutions.



Cluster profiles: mean standardised scores of the 3 clusters.

Figure 48. Cluster profiles: mean standardised scores of the three clusters.



Respondents on the first two principal components (83% of the variance), coloured by cluster.

Figure 49. The respondents on the first two principal components, coloured by cluster.

Choosing the number of clusters

SanAML compares solutions with 2 to 8 clusters.

Table 26

Solutions with 2 to 8 clusters for segments_demo.csv

Clusters	Average silhouette	Calinski-Harabasz	Gap (SE)	Smallest cluster
2	.437	323.7	0.701 (0.020)	40.3%
3 (shown)	.520	466.8	1.194 (0.019)	25.0%
4	.375	349.3	1.157 (0.025)	17.5%
5	.364	283.2	1.129 (0.023)	10.0%

Clusters	Average silhouette	Calinski-Harabasz	Gap (SE)	Smallest cluster
6	.166	243.6	1.101 (0.025)	12.5%
7	.169	220.8	1.099 (0.025)	10.0%
8	.170	200.1	1.057 (0.025)	9.4%

Concept: silhouette, gap statistic and stability

The *silhouette* of a customer compares how close the customer is to the own cluster with how close it is to the nearest other cluster; the average silhouette runs from -1 to 1. The *Calinski-Harabasz* index is the ratio of between-cluster to within-cluster spread. The *gap statistic* compares the within-cluster spread with what random data without groups would give; choose the smallest k whose gap is within one standard error of the next. *Stability* is the mean adjusted Rand index between the solution on all customers and the solutions on 20 random 80% subsamples (1 = identical groupings). The within-cluster sum of squares always falls as clusters are added, so look for an elbow.

Cut-off values: clusters

Average silhouette: .71–1.00 strong structure, .51–.70 reasonable, .26–.50 weak (possibly artificial), .25 or less no substantial structure (Kaufman & Rousseeuw, 1990). **Cluster size:** at least 5% of the respondents and at least 20. **Stability:** a mean adjusted Rand index of .80 or more (informal guide; Hubert & Arabie, 1985; Hennig, 2007).

What it means for our example

Silhouette (.520), Calinski-Harabasz (466.8), the gap statistic (1.194) and the elbow all point to three clusters. The three-cluster solution has reasonable structure (silhouette .52), a smallest cluster of 25.0% (90 shoppers) and perfect stability (mean adjusted Rand index 1.00 over 20 subsamples). With four clusters the silhouette falls to .37 (weak structure), the smallest cluster is 17.5% and stability is .92: the extra cluster splits a group in two similar parts instead of finding a new type.

Profiling and naming the clusters

Table 27

The three clusters of segments_demo.csv (means of the clustering variables; standardised means in brackets)

Variable	Cluster 1 (n = 144)	Cluster 2 (n = 126)	Cluster 3 (n = 90)
Price_sensitivity	5.97 (1.04)	3.42 (-0.58)	2.98 (-0.86)
Brand_loyalty	2.55 (-0.81)	6.12 (1.21)	3.28 (-0.40)
Quality_focus	3.16 (-1.03)	5.56 (0.62)	5.80 (0.78)
Online_affinity	4.74 (0.19)	3.03 (-1.04)	6.07 (1.15)
Social_influence	3.04 (-0.85)	4.37 (0.14)	5.77 (1.17)

Table 28

The clusters on variables that were not used to build them

Variable	Cluster 1	Cluster 2	Cluster 3
Age, mean (SD)	34.29 (8.57)	48.98 (10.22)	27.30 (5.47)
Income_k, mean (SD)	31.88 (8.87)	56.96 (15.62)	46.72 (13.65)
Channel: Store	28.5%	59.5%	8.9%
Channel: Online	20.1%	26.2%	60.0%
Channel: Mixed	51.4%	14.3%	31.1%
Gender: female	60.4%	55.6%	47.8%

What it means for our example

Cluster 1 (40%) is price sensitive and has low brand loyalty, quality focus and social influence; it is younger (mean age 34.29) and has the lowest income (31.88). Cluster 2 (35%) combines high brand loyalty and quality focus with low price sensitivity and low online affinity; it is the oldest group (48.98), has the highest income (56.96) and shops mostly in stores (59.5%). Cluster 3 (25%) is quality focused, online and socially influenced, with low price sensitivity; it is the youngest (27.30) and shops mostly online (60.0%). The differences in age (Welch $F(2, 234.1) = 202.04$, $p < .001$, $\eta^2 = .51$), income ($F(2, 191.2) = 143.69$, $p < .001$, $\eta^2 = .42$) and channel ($\chi^2(4) = 98.99$, $p < .001$, Cramér's $V = .37$) are large, whereas gender does not differ ($\chi^2(2) = 3.58$, $p = .167$, $V = .10$). Possible names, which are our suggestion and not a program result: *price-driven bargain hunters*, *loyal traditionalists* and *connected quality seekers*.

Tip: how to name a cluster

Use the standardised means: values above about +0.5 or below -0.5 characterise the cluster (SaNaML uses this limit in its summary). Name the cluster by its two or three most distinctive traits and check the names against the variables that were not used to build the clusters (age, income, channel), as above. Cluster numbers are only labels sorted by size; they can change when you change the data or the settings.

Hierarchical clustering with Ward's method (choose **Method** and click **Find the clusters** again) gives the same three clusters (sizes 40%, 35% and 25%, silhouette .52, stability .99), which is reassuring: well-separated groups do not depend on the algorithm.

How to report it

"K-means cluster analysis of Price_sensitivity, Brand_loyalty, Quality_focus, Online_affinity and Social_influence (standardised; $n = 360$) gave 3 clusters of 144, 126 and 90 respondents (k chosen by the largest average silhouette; average silhouette = .52, mean adjusted Rand index over 20 subsamples = 1.00)."

7.2 Segments in the customer file

Now cluster the customers of practice_marketing.csv on five behaviour variables.

1. On **Data**, open practice_marketing.csv. Click **Clustering**.
2. Click **Select none**, then tick Visits_per_month, Avg_basket, Discount_share, Satisfaction_1to10 and Tenure_months.
3. Under **Compare the clusters on other variables**, tick Region, Main_channel, Churned, Loyalty_member, Annual_spend and Complaints_12m.
4. Tick **Also compute the gap statistic (slower)** and click **Find the clusters**.

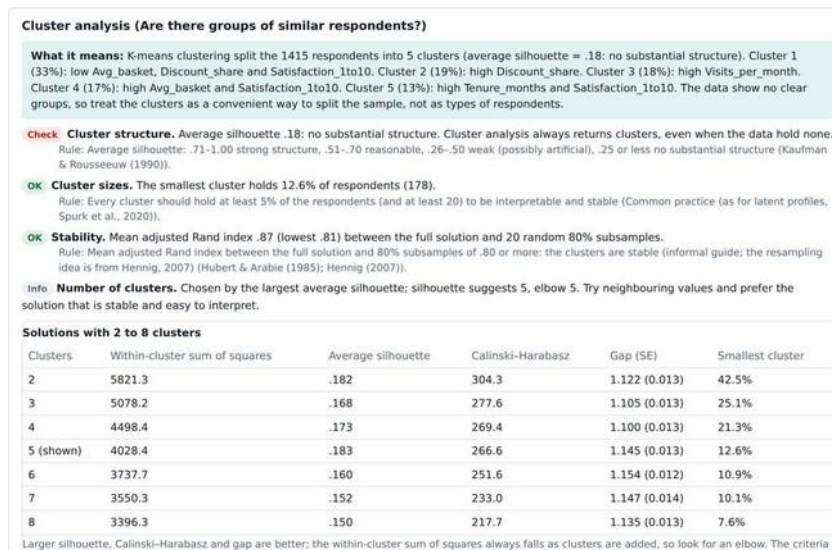


Figure 50. The report for the customer file: the summary, the checks and the table of solutions.

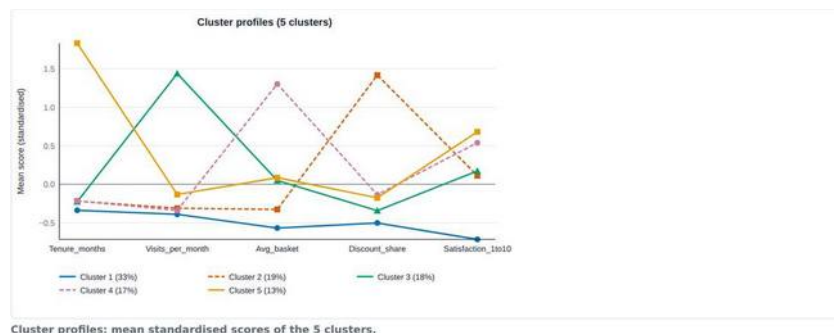


Figure 51. Cluster profiles of the five customer clusters.

Table 29

The five customer clusters (means; standardised means in brackets)

Variable	Cluster 1 (n = 461)	Cluster 2 (n = 271)	Cluster 3 (n = 258)	Cluster 4 (n = 247)	Cluster 5 (n = 178)
Tenure_months	23.63 (−0.34)	26.18 (−0.22)	26.00 (−0.23)	26.25 (−0.21)	69.94 (1.83)
Visits_per_month	2.48 (−0.39)	2.64 (−0.31)	6.06 (1.44)	2.58 (−0.34)	2.99 (−0.13)
Avg_basket	79.77 (−0.57)	85.57 (−0.33)	94.58 (0.05)	124.69 (1.30)	95.45 (0.09)
Discount_share	0.19 (−0.50)	0.47 (1.42)	0.21 (−0.34)	0.24 (−0.14)	0.24 (−0.18)
Satisfaction_1to10	5.51 (−0.71)	6.82 (0.11)	6.91 (0.17)	7.49 (0.54)	7.72 (0.68)

What it means for our example

SaNaML chose five clusters (largest silhouette) from 1415 customers (85 with a missing satisfaction rating were left out), but the average silhouette is only .18: "no substantial structure". The silhouette hardly changes between 2 and 8 clusters (.150 to .183), and the clusters are stable (mean adjusted Rand index .87). The data therefore form a continuous cloud, not distinct types. The five clusters can still be used as a *descriptive* grouping of customers, as SaNaML's own summary says: "treat the clusters as a convenient way to split the sample, not as types of respondents". They are: a large, less satisfied, low-basket group (Cluster 1, 33%); discount shoppers (Cluster 2, 19%, discount share 0.47); frequent visitors (Cluster 3, 18%, 6.06 visits per month); big-basket customers (Cluster 4, 17%, basket 124.69); and long-tenure satisfied customers (Cluster 5, 13%, 69.94 months).

Note: stable does not mean real

High stability (.87) tells you that the same customers fall into the same clusters when the sample is perturbed. It does not tell you that the groups exist in nature: a method that cuts a continuous cloud into pieces can do so repeatedly. Read the silhouette first.

7.3 Keep the clusters and compare them

The second card on the Clustering screen, 2 · *Keep the clusters (Use them in other analyses)*, adds each customer's cluster to the data as a grouping variable.

1. In **Name of the new variable** type **Segment**.
2. Click **Add to the data**. SaNaML confirms: "Added "Segment"."
3. Use Segment in any analysis. Customers who were left out of the clustering have no cluster.

Churn by segment. Choose **Analyses** → **Chi-square, Fisher's exact and McNemar tests** with Segment and Churned (Figure 52).

Chi-square test of independence (Segment by Churned)

What it means: Segment and Churned are related (Cramér's $V = .20$, small). The cells with large adjusted residuals (beyond ± 1.96) show where the pattern comes from.

OK Expected counts. All expected counts are adequate (smallest 29.44).

Rule: Chi-square: no expected count below 1 and no more than 20% below 5; otherwise use the exact test (Cochran (1954)).

Info Effect size. Cramér's $V = .20$ (small).

Rule: Cramér's V : small, medium and large are .10, .30 and .50 divided by $\sqrt{(\text{smaller table dimension} - 1)}$ (Cohen (1988)).

Counts (row %)			
Segment	No	Yes	Total
Cluster 1	341 (74%)	120 (26%)	461
Cluster 2	222 (82%)	49 (18%)	271
Cluster 3	231 (90%)	27 (10%)	258
Cluster 4	222 (90%)	25 (10%)	247
Cluster 5	165 (93%)	13 (7%)	178

Tests			
Test	Value	df	p
Pearson chi-square	55.82	4	< .001
Likelihood ratio	56.43	4	< .001

Adjusted standardised residuals			
Segment	No	Yes	
Cluster 1	-6.68 *	6.68 *	

Figure 52. Churned by Segment (chi-square test of independence).

Table 30

Churned by customer segment

Segment	No	Yes	Total
Cluster 1	341 (74%)	120 (26%)	461
Cluster 2	222 (82%)	49 (18%)	271
Cluster 3	231 (90%)	27 (10%)	258
Cluster 4	222 (90%)	25 (10%)	247
Cluster 5	165 (93%)	13 (7%)	178

What it means for our example

Churn differs between the segments, $\chi^2(4, N = 1415) = 55.82$, $p < .001$, Cramér's $V = .20$. Cluster 1 (the less satisfied group) churns more than expected (adjusted residual +6.68, 26% against 7% in Cluster 5). Because satisfaction helped to build the clusters, and satisfaction also predicts churn (chapter 6), this result is partly expected. It nevertheless shows the practical value of the segments: one customer in four in the largest segment left, and one in fourteen in the long-tenure segment.

Region by segment. Replace Churned by Region: the segments do not differ by region, $\chi^2(16, N = 1415) = 18.40$, $p = .301$, Cramér's $V = .06$. A segmentation that does not depend on region can be used in all regions.

Spend by segment. A one-way ANOVA of Annual_spend by Segment shows a large difference, Welch's $F(4, 565.77) = 212.94$, $p < .001$, $\omega^2 = .403$, with the frequent visitors (Cluster 3) spending most (mean 6407.81 against 2240.58 in Cluster 1).

Note: a segment comparison on a variable that built the segment

Cluster 3 was defined by frequent visits, and annual spend is largely visits times basket (chapter 5). The spend difference is therefore built in, not an independent finding. SaNaML says the same: tests on variables that were *not* used to form the clusters are meaningful, but the clusters were chosen from the same sample, so confirm in new data before calling them types. Region and Churned were not used here, so those comparisons are fair; Annual_spend is a consequence of the clustering variables.

Tip: profile with the right variables

Cluster on the variables that you can act on (behaviour, attitudes) and profile with variables that tell you how to reach the clusters (region, channel, age) and with business outcomes (churn, spend). Keep clustering variables and profiling variables apart.

Note: protect small groups and individuals

Segments describe patterns in groups. Do not use them as decisions about individual customers without checking the consequences, and keep very small groups private when you report results.

Exercises

Exercise: segmentation

1. Cluster segments_demo.csv on the five scales with **Compare solutions from 2 to 8**, and ask for four clusters (**Show 4 clusters**). Which statistic tells you that the fourth cluster is not needed? Answer: the silhouette drops from .520 to .375 and the gap statistic does not rise.
1. Cluster practice_marketing.csv on Visits_per_month and Avg_basket only. Does the silhouette improve? Report what you see.
1. Planted answer for the chapter: three segments in segments_demo.csv (silhouette .52, sizes 40%, 35%, 25%); no clear structure in the customer file (silhouette .18).

8. Putting it together: a worked mini-report

The earlier chapters each ended with one result. This chapter shows how to turn them into a short report that you can hand to a colleague, using the Word report of SaNaML.

8.1 The question and the plan

Imagine a retailer that tested a new offer and wants to know three things: did the new offer work, who is at risk of leaving, and which kinds of customers does it have? A plan with the tools of this manual:

Table 31

From marketing questions to SaNaML screens

Question	Analysis	Section
Did the New offer raise the response rate?	Chi-square test of Responded by Campaign_group	4.1
Did it raise spend more than the Standard offer?	Independent-samples t-test of Spend_change by Campaign_group	4.3
Who is likely to leave, and why?	Logistic regression of Churned; machine learning comparison; explanations; calibration	6.2 to 6.7
Which kinds of customers do we have?	Clustering on behaviour; churn and region by segment	7.2 and 7.3
How many customers does the next test need?	A priori power analysis for a chi-square test	4.4

8.2 Collect the results

Every result has an **Add to Word report** button under its *How to report it* sentence. Press it for each result you want to keep. The menu item **Word report** then shows the number of results, for example **Word report (4)**.

For the example below we added four results: the chi-square test of Responded by Campaign_group, the t-test of Spend_change by Campaign_group, the logistic regression of Churned (four predictors) and the box plot of Annual_spend by Loyalty_member.

8.3 The Word report screen

Click **Word report** (Figure 53). The screen lists the results in the order added, each with its title and its *What it means* sentence, and buttons to move a result up (↑) or down (↓) or remove it. Put the results in the order of your argument. Type a **Report title** and click **Download Word report (.docx)**.

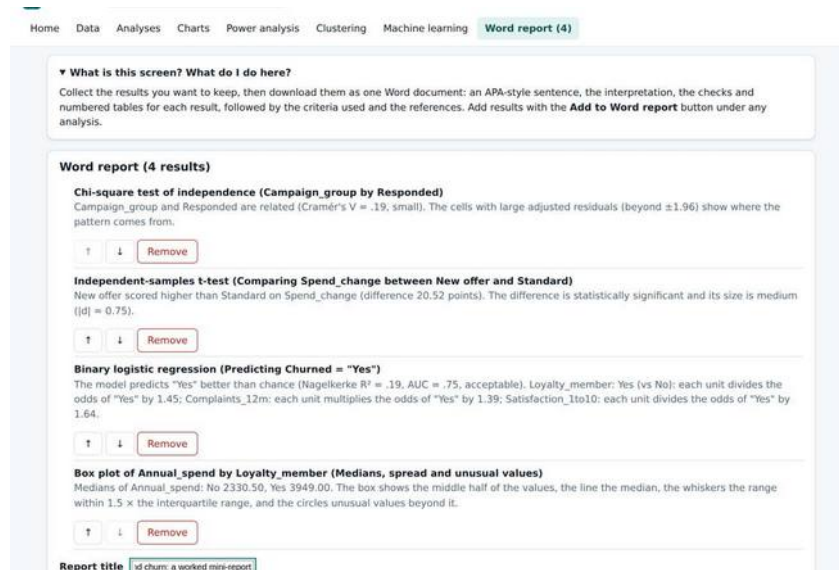


Figure 53. The Word report screen with four results and the report title.

What it means for our example

The downloaded document is titled "Campaign and churn: a worked mini-report" and states: "Analyses run in SaNaML 0.10.2 on 2026-10-06. All computations were done on the analyst's own computer." For each result it gives the APA-style sentence, the interpretation, the checks and numbered tables (the chi-square test alone gives four tables: counts, tests, adjusted residuals and effect sizes), then the criteria used and the references. Figures are included.

Note: the document is a draft

The report documents the analyses; it does not interpret them for your audience. Add the context, the limitations and the recommendations in Word. In particular state which customers were left out of each analysis, that the practice data are simulated (if you use them), and which variables were chosen on theory and which by the data.

8.4 A model paragraph

The paragraph below is built only from numbers of this manual.

How to report it

"The new offer raised the response rate from 26% (199 of 762 customers) to 44% (327 of 738), $\chi^2(1, N = 1500) = 54.50$, $p < .001$, Cramér's $V = .19$. Monthly spend rose by 32.13 after the new offer and by 11.61 after the standard offer, Welch's $t(863.00) = 14.33$, $p < .001$, $d = 0.75$. Churn (16.7% of customers) was predicted by low satisfaction (OR = 0.61 per point), more complaints (OR = 1.39 per complaint) and non-membership of the loyalty programme (OR for members = 0.69); the four-predictor model reached AUC = .754 in the sample, and in cross-validation on customers not used to fit it a logistic regression with 13 predictors reached AUC = .755. Flexible machine learning models did not outperform logistic regression. On cross-validated predictions the logistic regression at a cut-off of 0.50 found 13% of the leavers (precision 0.50), whereas at a cut-off of 0.20 it found 60% of them (precision 0.33), so the cut-off for a retention campaign should be chosen from the costs of contact and of losing a customer."

(The AUC of .754 belongs to the logistic regression of section 6.2, fitted to the 1415 customers with a satisfaction rating. The AUC of .755 is the average over the five test folds of the machine learning comparison, which used 13 predictors and all 1500 customers.)

8.5 A checklist before you send the report

- State the **sample**: how many customers, how many were left out of each analysis, and why.
- Report **effect sizes and intervals**, not only p-values, and the rates behind a test (26% against 44%).
- State the **checks** that failed and what you did (for example Welch's test because the variances differed; the logarithm because of skew; zero-spend customers excluded from the logarithmic model).
- For predictions, report **performance on unseen customers** and the **baseline** (the null model), do not report accuracy alone for a rare outcome, and state the **cut-off** with the recall and precision it gives.
- Say that explanations (SHAP, ALE) describe the model, **not causes**.
- Describe **segments** as descriptions of this sample, with the silhouette and the stability.
- If the data are simulated or a teaching file, say so.

Exercise

Exercise: your own mini-report

Reproduce the four results of section 8.2, add the cluster analysis of section 7.2 and the t-test of section 5.4, put them in a sensible order, and write two sentences of recommendation under each result in Word.

Appendix A. Codebooks

All three files are simulated. Missing values are blank cells.

Table 32

practice_marketing.csv (1,500 customers)

Variable	Meaning
CustomerID	Identifier (a label; do not use it as a predictor)
Age, Gender, Region	Customer profile; Gender is female, male or other; Region is Central, East, North, South or West
Income_k	Household income in thousands (81 missing; right-skewed)
Tenure_months	Months as a customer
Loyalty_member	Yes or No
Main_channel	Store, Online or Mobile app
Visits_per_month, Avg_basket, Discount_share	Shopping behaviour (visits per month, average basket value, share of spend bought on discount)
Complaints_12m	Count of complaints in the last 12 months
Satisfaction_1to10, Recommend_0to10	Survey ratings (85 and 65 missing)
Annual_spend	Spend in the year (right-skewed; 82 customers spent 0)
Campaign_group	Standard (762) or New offer (738), randomly assigned
Responded	Yes or No: responded to the campaign (526 Yes)
Monthly_spend_before, Monthly_spend_after	Monthly spend around the campaign (paired)
Churned	Yes or No: left within the year (250 Yes, 16.7%)

Table 33

What was planted in practice_marketing.csv, and where this manual finds it

Planted effect	Result in this manual	Section
New offer responds more than Standard (about 44% and 26%)	44% against 26%; $\chi^2(1) = 54.50$; $V = .19$	4.1
Monthly spend rises after the campaign	Paired $t(1499) = 28.78$; $d_z = 0.74$	4.2
Annual_spend depends on income, loyalty, satisfaction, visits	$R^2 = .806$ (raw), .415 (Annual_spend_ln1p), .773 (Annual_spend_ln)	5.1, 5.2
Loyalty members spend more	$d = -0.79$, medians 3949.00 against 2330.50	5.4
Churn depends on satisfaction, complaints, tenure and loyalty	OR 0.61, 1.39, 0.992, 0.69; AUC .754	6.2
Response depends on the offer, loyalty and channel	Section 6.8	6.8
Customers can be segmented on behaviour	Silhouette .18: no substantial structure	7.2

Table 34*practice_brand_survey.csv (500 respondents)*

Variable	Meaning
RespondentID	Identifier
Age_group, Gender, Customer_type	18-24, 25-34, 35-49 or 50+; female or male; Long-term, New or Regular
Trust1 to Trust4	Trust items, 1 to 7
Value1 to Value4	Perceived value items, 1 to 7 (Value2 has 18 missing)
Loyalty1 to Loyalty3	Loyalty items, 1 to 7 (Loyalty2 has 18 missing)

Ten respondents answered every item identically (straight-lining); see section 3.1.

Table 35*segments_demo.csv (360 shoppers)*

Variable	Meaning
ID	Identifier
Price_sensitivity, Brand_loyalty, Quality_focus, Online_affinity, Social_influence	Attitude scores, 1 to 7
Age, Income_k, Gender, Channel	Profile; Channel is Mixed, Online or Store

Three segments were planted; see section 7.1.

Appendix B. Marketing question to tool

Table 36

Where to find each marketing question

Marketing question	Tool in SaNaML	Section
Who are my customers? How are spend, tenure and satisfaction distributed?	Analyses → Descriptive statistics; Charts → Histogram, Box plot	2.1 to 2.3
Do groups differ on a measure? (channels, regions, members)	Charts → Means with confidence intervals; Analyses → t-test, One-way ANOVA	2.4, 5.4, 5.5
Which measures go together?	Charts → Correlation heatmap, Scatter plot; Analyses → correlation	2.5
How are customers split across categories?	Charts → Bar chart of percentages	2.6
How do I take the logarithm of spend, or make low and high groups, a 0/1 indicator or merged categories?	Data → 4 · Compute and recode variables	2.6, 2.7
Is my survey scale reliable? Is someone straight-lining?	Data → Build construct scores; Data quality screening	3.1, 3.2
What drives loyalty or satisfaction?	Regression of the scale score	3.4
Did the offer change the response rate?	Chi-square test (A/B test)	4.1
Did spend change after a campaign?	Paired t-test (and Wilcoxon); t-test of the change between groups	4.2, 4.3
How many customers does a test need?	Power analysis	4.4
What drives spend?	Multiple linear regression; hierarchical regression	5.1 to 5.3
Which customers are likely to churn or respond?	Logistic regression; Machine learning	6.2 to 6.4
What drives the prediction?	Machine learning → Explain this model (SHAP, ALE)	6.6
Can the predicted probabilities be trusted?	Machine learning → Calibration	6.7
What happens when probabilities become yes or no? How many flagged customers really leave?	Machine learning → Cut-offs, precision and recall	6.5
Which kinds of customers do I have?	Clustering; keep the clusters; compare on churn and region	7.1 to 7.3
How do I keep my results?	Add to Word report → Word report	8.2, 8.3

Appendix C. Glossary

Table 37

Terms used in this manual

Term	Meaning
A/B test	A comparison of two versions of an offer given at random to comparable customers
ALE (accumulated local effects)	How a model's prediction changes across the range of one predictor, keeping the data realistic
AUC	Area under the ROC curve: the chance that a customer who left gets a higher predicted risk than one who stayed
Balanced accuracy	The average of recall and specificity
Calibration	Whether predicted probabilities match how often the event happens
Churn	A customer leaving within the observation period
Cohen's d, d _z	Standardised difference between two means (d _z : for paired measurements)
Cramér's V, w	Effect size for a table of counts; for a 2 × 2 table equal to ϕ and to Cohen's w
Cronbach's α	A measure of the internal consistency of a scale
Cut-off (threshold)	The predicted probability above which a customer is called a case (for example a leaver)
Cross-validation	Testing a model on customers it was not fitted on, by splitting the sample into folds
Imbalanced outcome	An outcome in which one answer is much rarer than the other (here churn, 16.7%)
Indicator variable	A 0/1 variable that is 1 for the chosen category (section 2.7)
Leakage	A predictor that contains or is a consequence of the outcome
Median split	Two groups made at the median of a scale variable; a value equal to the cut point goes to the lower group
Null model	A model that predicts the average (or the most common answer) for everybody
Odds ratio	The factor by which the odds of an event are multiplied for one more unit of a predictor
Precision	Of the customers flagged, the share who really have the outcome
Recall (sensitivity)	Of the customers who really have the outcome, the share that was flagged
SHAP	A method that splits a prediction into the contributions of the predictors
Silhouette	A measure of how well a customer fits its own cluster compared with the nearest other cluster
Straight-lining	Giving the same answer to all items of a questionnaire
Stability (clusters)	Agreement between the clusters on all customers and on random subsamples (adjusted Rand index)
Welch's test	A test of means that stays valid when the group variances differ

Appendix D. Limits and doubtful points to keep in mind

Table 38

Things to check when you work with these tools

What you see	What to do
A ratio or percentage change is missing for some customers	A customer whose B value is 0 gets no value (82 customers have a Monthly_spend_before of 0, so the percentage change exists for 1418); say how many customers were left out (section 4.3)
After a variable is added in step 4, the message "'...' cannot be used" appears under the name box	The suggested name is now taken by the variable you just added; the variable is in the list. Type another name to make a further one
A median split of a scale variable	It discards information; keep the scale variable for the analysis and use the groups for reporting (section 2.7)
The even-odd consistency index is listed as "Not used"	It needs at least four scales with two or more items; with fewer, rely on the other indices and the rule of two flags
The sign of d in a t-test depends on the order of the groups	SaNaML reports d and explains the sign under the effect size; read the group means and say which group is higher
The 0.5 cut-off of the logistic regression's classification table	Look at other cut-offs with the machine learning comparison (section 6.5)
Machine learning is slow	Use the Quick look while exploring, untick slow models and unneeded predictors
My results are gone	Results are kept per browser; download the Word report at the end of each session

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